

TEMPLATE

KEY PROJECT INFORMATION & PROJECT DESIGN DOCUMENT (PDD)

PUBLICATION DATE **14.10.2020**

VERSION v. **1.2**

RELATED SUPPORT

– **TEMPLATE GUIDE Key Project Information & Project Design Document v.1.2**

This document contains the following Sections

Key Project Information

Section A - Description of Project

Section B - Application of approved Gold Standard Methodology (ies) and/or demonstration of SDG Contributions

Section C – Duration and crediting period

Section D – Summary of Safeguarding Principles and Gender Sensitive Assessment

Section E – Outcome of Stakeholder Consultations

Appendix 1 – Safeguarding Principles Assessment (mandatory)

Appendix 2 - Contact information of Project participants (mandatory)

Appendix 3 - LUF Additional Information (project specific)

Appendix 4 - Summary of Approved Design Changes (project specific)

KEY PROJECT INFORMATION

GS ID of Project	GS11207
Title of Project	Multi-Layer Household Water Filtration System in Kenya
Time of First Submission Date	27/05/2021
Date of Design Certification	17/11/2021
Version number of the PDD	06
Completion date of version	10/02/2022
Project Developer	Profit Carbon Environmental Energy Technology (Shanghai) Co., Ltd.
Project Representative	The official focal point: Profit Carbon Environmental Energy Technology (Shanghai) Co., Ltd.
Project Participants and any communities involved	The project owner: Climate Neutral Kenya Limited
Host Country (ies)	Republic of Kenya
Activity Requirements applied	☒ Community Services Activities ☐ Renewable Energy Activities ☐ Land Use and Forestry Activities/Risks & Capacities ☐ N/A
Scale of the project activity	☐ Micro scale ☐ Small Scale ☒ Large Scale
Other Requirements applied	N/A
Methodology (ies) applied and version number	<i>Methodology for Emission Reductions from Safe Drinking Water Supply</i> (Version 1.0)
Product Requirements applied	☒ GHG Emissions Reduction & Sequestration ☐ Renewable Energy Label ☐ N/A
Project Cycle:	☐ Regular ☒ Retroactive

Table 1 – Estimated Sustainable Development Contributions

Sustainable Development Goals Targeted	SDG Impact (defined in B.6.)	Estimated Annual Average	Units or Products
6 Clean Water and Sanitation	Total number of person.days served with satisfactory level of safe water	692,898,179	Person.days
7 Affordable and Clean Energy	Number of water purifiers sold each year	128,970	Units
8 Decent Work and Economic Growth	Number of males and females employed by the project	12 jobs created including 6 males and 6 females	
8 Decent Work and Economic Growth	Percentage of employees with salaries paid at par with the average wage of Kenya	100	%
13 Climate Action	GHG emission reductions	459,638	tCO ₂ e

SECTION A. DESCRIPTION OF PROJECT

A.1. Purpose and general description of project

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The Bill of Rights of the Constitution in article 43 (d) states that every person has right to clean and safe water in adequate quantities. However, with a population of 50 million, 32% of Kenyans rely on unimproved water sources for drinking, such as ponds, shallow wells and rivers, which are often turbid and thus pose a great health risk to these families. Unsafe drinking water and sanitation is the second leading risk factor and contributor to all mortality (deaths) and morbidity burden (DALY) in Kenya¹. The challenge is especially evident in rural areas and urban slums where people are often unable to connect to piped water infrastructure².

For drinking and cooking, most of the local households' boil water to purify it³. Woody biomass is the main energy source in Kenya since most Kenyan households use fuelwood and/or charcoal

¹ Kenya Environmental Sanitation and Hygiene Policy 2016-2030. Ministry of Health. <https://www.wsp.org/sites/wsp.org/files/publications/Kenya%20Environmental%20Sanitation%20and%20Hygiene%20Policy.pdf>

² <https://water.org/our-impact/where-we-work/kenya/>

³ <https://www.iamat.org/country/kenya/food-and-water-safety>

for heating needs ⁴. It is expected that major reliance of woody fuels will continue for decades, in the face of dwindling feedstock supply due to severe forest loss. From 2002 to 2019, Kenya lost 45.8kha of humid primary forest, making up 14% of its total tree cover loss in the same time period⁵. Furthermore, there are a fraction of households even cannot afford firewood to boil water, hence are forced to continue to drink unsafe water.

Sustainable use and management of water resources is key to the successful implementation of the Country's development agenda and achieving sustainable development goals. The Ministry will increase percentage of national population with access to safe water from 60% in 2017 to 80% in 2022. Consequently, urban water coverage will increase from 70.3% to 83% and water coverage in rural areas will be increased from 55.9% to 78%⁶.

The Multi-Layer Household Water Filtration System in Kenya project is invested and implemented by Climate Neutral Kenya Limited⁷ in Makueni, Trans Nzoia, Kirinyaga County, Bungoma and Nakuru County of Republic of Kenya. The multi-Layer water purifiers involved in this project are produced by Zhejiang Duanqi Renewable Energy Co., Ltd. The sales are conducted by TRIOPT AFRICA LIMITED⁸ and the after-sales service are provided by Climate Neutral Kenya Limited. The water purifiers in this project are sold at a low market price⁹, in order to keep the project running normally, A whole set of the alternative dome ceramic, carbon cartridge and mineral stone will be supplied to the end-users together with the water purifiers. All the alternative filtration system can be changed easily and will not affect the usage of the water purifiers on that day. Considering

⁴ <https://www.sciencedirect.com/science/article/pii/S0961953420300532>

⁶ Strategic Plan 2018-2022. Ministry of Water and Sanitation. <https://africacheck.org/wp-content/uploads/2019/04/Ministry-of-Water-Kenya-Strategic-Plan-2018-2022-Final-Version-Dec2018.pdf>

⁷ Climate Neutral Kenya Limited is on 9 Oct 2020 Incorporated under the Companies Act, 2015 and that the Company is a PRIVATE LIMITED COMPANY

⁸ TRIOPT AFRICA LIMITED is a subsidiary of TAL HOLDING LIMITED (one of the funders of the project owner CNKL), so TRIOPT AFRICA LIMITED and CNKL has the same investors- TAL HOLDING LIMITED.

⁹ As per “sales contract” between project owner and users, the sold price of the water purifier (water output capacity is 16L) involved in this project is 2,350 KES, and the project owner promises that the price will be the highest selling price, and the future selling price can only be lower than this price, not higher than this price. According to public information (<https://www.saset.co.ke/water-purification-machine-cost-in-kenya/>), A good water purifier for a household cost anywhere between KSH 100,000 to KSH 150,000.

Also, in the same type of water purifier, the price of 2,350 KES is also a lower market price(https://jiji.co.ke/nairobi-central/kitchen-appliances/water-filter-xvt47PQkT10deFbNLqUUR7h4.html?lid=KXVQmkWCY2M9OQOq&cur_pos=2&pos=2&ads_count=621&ads_per_page=17&page=1).

carbon revenue of the project, all the alternative materials needed will be provided by the project owner for free.

The project involves introduction of multi-Layer water purifiers as a way to provide safe drinking water for households in Kenya. Households simply pour untreated water into the water purifier. Water seeps through 8 step filtration system, removing bacteria and other suspended solids, and treated water is stored in the lower tank, which is easily accessed by turning on the faucet. The water filtration system requires no energy input or consumables. With a capacity to purify 3 liters of water per hour over a 15-year lifespan of the water purifiers, the project technology can provide enough water to meet and exceed the World Health Organization recommended domestic water consumption of 7.5 liters per capita per day¹⁰.

The water purifiers of the project will offer an affordable, long-term and zero emission solution for households that generally consume unsafe drinking water. It will not only dramatically increase access to safe drinking water but also reduce consumption for woody fuels previously required to treat drinking water, which will decrease environmental degradation and greenhouse gas emissions.

In order to cover more households with high-quality water purifiers at a lower price, it is restricted in the project that each household can only buy one set of water purifier, the Specific sales plan in the first crediting period is shown in the table below:

Year	Water purifiers sold that year (Sets)	Accumulated number of water purifiers sold (Sets)
1	480,000	480,000
2	122,500	602,500
3	32,500	635,000
4	7,500	642,500
5	2,350	644,850

It is estimated that about 644,850 water purifiers will be sold and about 2.5149 million¹¹ people would have access to clean water during the first crediting period. 459,638 tCO₂e emission reductions will be produced annually and the total emission reductions during the first crediting period is 2,298,193 tCO₂e.

A.1.1 Eligibility of the project under Gold Standard

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¹⁰ https://www.who.int/water_sanitation_health/emergencies/qa/emergencies_qa5/en/

¹¹ 644,850 (water purifiers sold) * 3.9 (average person per household) = 2.5149 million.

The proposed project falls under section 3.1.1 of *GS4GG Principles & Requirements* (Version 1.2), section 3 of *GS4GG Community Services Activity Requirements* (Version 1.2) and section 2 of *GS4GG GHG Emissions Reduction & Sequestration Product Requirements* (Version 2.0) with the following eligibility criteria:

Eligibility Criteria	Justification
(a) Types of Projects Section 3.1.1 of <i>GS4GG Principles & Requirements</i> (Version 1.2) Eligible projects shall include physical action/implementation on the ground. Pre-identified eligible project types are identified in the Eligibility Principles and Requirements section. Section 3.1.1 of <i>GS4GG Community Services Activity Requirements</i> (Version 1.2) Pre-identified CSA project types are a) Renewable energy; b) End-use energy efficiency; c) Waste management and handling; d) Water, sanitation and hygiene (WASH). Section 5.1.1 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements</i> (Version 2.0) The Following Project types are eligible for issuance of Gold Standard VERs: a) Renewable Energy Supply; b) End-Use Energy Efficiency Improvement; c) Waste Handling & Disposal; d) Land Use and Forests.	The project involves introduction of multi-Layer water purifiers as a way to provide safe drinking water for households in Makueni County, Trans Nzoia County, Kirinyaga County, Bungoma County and Nakuru County of Kenya. It will not only dramatically increase access to safe drinking water but also reduce consumption for woody fuels previously required to treat drinking water. Therefore, the project activity falls under section 3.1.1 (b) "End-Use Energy Efficiency Improvement: Project activities that reduce energy requirements as compared to baseline scenario without affecting the level and quality of services or products, where the end user of the products and services are clearly identified and when the physical intervention is required at the user end." and (d) "Water, sanitation and hygiene (WASH): WASH activities contributing to climate change mitigation and/or adaptation benefits." of <i>GS4GG Community Services Activity Requirements</i> (Version 1.2). Besides, the project activity falls under section 5.1.1 (b) "End-Use Energy Efficiency Improvement: Project activities that reduce energy requirements as compared to baseline scenario without affecting the level and quality of services or products, where the end user of the products and services are clearly identified and when the physical intervention is required at the user end." of <i>GS4GG GHG</i>

	<i>Emissions Reduction & Sequestration Product Requirements (Version 2.0).</i> As per section 4.1.3 of <i>GS4GG Principles & Requirements (Version 1.2)</i>, "A Project type is automatically eligible for Gold Standard Certification if there are Gold Standard approved Activity Requirements and/or Impact Quantification Methodologies associated with it or it's referenced in the Gold Standard Product Requirements". The project type is automatically eligible for Gold Standard Certification.
(b) Location of Project: Section 3.1.1 of <i>GS4GG Principles & Requirements (Version 1.2)</i> Projects may be located in any part of the world. Section 3.1.2 of <i>GS4GG Community Services Activity Requirements (Version 1.2)</i> Project Area and Boundary shall be defined in line with the applicable Impact Quantification Methodologies and Product Requirements. Section 3.1.1 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements (Version 2.0)</i> Gold Standard VER Projects may be located in any host country or state. However, where host countries or states have mandatory operational schemes to reduce GHG emissions in any form (e.g. cap & trade, carbon tax etc.), Projects shall only be eligible if the Project Developer has either: (a) provided Gold Standard with satisfactory justification that no double counting of emission reductions occur or (b) has committed to retiring eligible units equal to the quantity of Gold Standard VERs. Refer to Annex A of this document.	The project location is the Republic of Kenya (refer to section A.2 of the PDD for detailed project area and boundary) and there are no mandatory operational schemes to reduce GHG emissions in any form (e.g., cap & trade, carbon tax etc.) in Kenya, which based on section 3.1.1 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements (Version 2.0)</i> is an eligible host country.
(c) Project Area, Project Boundary and Scale:	The project location is the Republic of Kenya (refer to section A.2 of the PDD for detailed project area and boundary) and there are no

Section 3.1.1 of <i>GS4GG Principles & Requirements</i> (Version 1.2) The Project Area and Project Boundary shall be defined. Projects may be developed at any scale although certain rules, requirements and limitations may apply under specific Activity Requirements, Impact Quantification Methodologies and Products Requirements. In order to avoid double counting the Project shall not be included in any other voluntary or compliance standards programme unless approved by Gold Standard (for example through dual certification). Also, if the Project Area overlaps with that of another Gold Standard or other voluntary or compliance standard programme of a similar nature, the project shall demonstrate that there is no double counting of impacts at design and performance certification (for example use of similar technology or practices through which the potential arises for double counting or misestimation of impacts amongst projects). Section 3.1.2 of <i>GS4GG Community Services Activity Requirements</i> (Version 1.2) Project Area and Boundary shall be defined in line with the applicable Impact Quantification Methodologies and Product Requirements. The definition of scale is the same for all Projects, except Microscale. Section 3.1.1 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements</i> (Version 2.0) Gold Standard VER Projects may be located in any host country or state. However, where host countries or states have mandatory operational schemes to reduce GHG emissions in any form (e.g., cap & trade, carbon tax etc.), Projects shall only be eligible if the Project Developer has either:	mandatory operational schemes to reduce GHG emissions in any form (e.g., cap & trade, carbon tax etc.) in Kenya, which based on section 3.1.1 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements</i> (Version 2.0) is an eligible host country. The project boundary is defined based on <i>Methodology for Emission Reductions from Safe Drinking Water Supply</i> (Version 1.0) and is limited to the Republic of Kenya (refer to section B.3 of the PDD). The project activity will use zero energy water purification technologies that do not deliver thermal energy. According to section B.4 of the PDD, the baseline scenario of the project is the practice of treating water for drinking by boiling it on stoves using non-renewable biomass. As per section 3.6.2 of the applied GS methodology, the amount of energy required to boil 1 L of water is 360.83 kJ/L¹². The estimated annual average quantity of safe water provided by the project is 564,888,600 L¹³. Based on the results of baseline survey, the weighted average efficiency of the baseline stove for water boiling is 16.1%¹⁴. Annual average thermal energy savings is $360.83 \text{ kJ/L} * 564,888,600 \text{ L} / 16.1\% / 1,000,000 = 1,267,249.928 \text{ GJ} = 1,056.042 \text{ GWh}_{\text{th}}$¹⁵, which is more than 180 GWh_{th}. As per section 9.1.1 and 9.1.2 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements</i> (Version 2.0), it's a large-scale GS VER project. Each water purifier will have a unique serial number, which will prevent any retailers or distributors from selling the same water purifier and registering it for two or more project developers. Sales records and end-user's database were kept for cross-check.
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¹² This is calculated from the specific heat of water of 4.186 kJ/L·°C, the difference between the initial and final water temperature assuming a start at 20 °C and end at 100 °C, evaporation of 1% of water required initially to obtain 1 L boiled water, and latent heat of water evaporation of 2,260 kJ/L ($4.186 * (100 - 20) * (1 + 1\%) + 1\% * 2,260$).

¹³ Refer to the ER Calculation Sheet for more details.

¹⁴ Refer to the ER Calculation Sheet for more details.

¹⁵ $1 \text{ GJ} = 3 / (3.6 * 10^3) = 8.3333 * 10^{-4} \text{ GWh}_{\text{th}}$

(a) provided Gold Standard with satisfactory justification that no double counting of emission reductions occurs or (b) has committed to retiring eligible units equal to the quantity of Gold Standard VERs. Refer to Annex A of this document. Section 9.1.1 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements (Version 2.0)</i> GSVER Projects may be registered as 'large scale', 'small scale' (for the applicability of methodologies and tools only) or 'microscale'. Scale is defined in the relevant Activity Requirements or where these do not exist then per following paragraphs. Section 9.1.2 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements (Version 2.0)</i> All Projects exceeding the small-scale thresholds are defined as large scale. Small scale projects are defined in accordance with CDM project standard for project activities, as below; (a) Type 1: Renewable energy Projects: maximum output capacity of 15 MW(e) or 45MW(th). (b) Type 2: End-use energy efficiency project improvement: activities that reduce energy consumption, on the supply and/or demand side, with a maximum energy saving of 60 GWh per year (or an appropriate equivalent) in any year of the crediting period. In this context, for project activities that improve thermal energy efficiency, the maximum energy saving of 60 GWh(e) per year is equivalent to 180 GWh(th) per year saving; (c) Type 3: Other project activities: project involves technologies such Safe Water Supply, Waste management, etc. not included in Type I or Type II that result in GHG emission reductions not exceeding 60,000 ton CO₂e per year in any year of the crediting period.	The project proponents will check on Gold Standard, CDM, VCS and other voluntary or compliance standard programme based on the repeated serial numbers, similar technology and project location. Declaration of No Double Counting by the project owner was provided as evidence. Therefore, the project area does not overlap with that of another Gold Standard or other voluntary or compliance standard programme of a similar nature.
(d) Host Country Requirements Section 3.1.1 of <i>GS4GG Principles & Requirements (Version 1.2)</i>	The project location is the Republic of Kenya (refer to section A.2 of the PDD for detailed project area and boundary) and there are no mandatory operational schemes to reduce

Projects shall be in compliance with applicable Host Country's legal, environmental, ecological and social regulations. Section 3.1.2 of <i>GS4GG Community Services Activity Requirements</i> (Version 1.2) Project Area and Boundary shall be defined in line with the applicable Impact Quantification Methodologies and Product Requirements. Section 3.1.1 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements</i> (Version 2.0) Gold Standard VER Projects may be located in any host country or state. However, where host countries or states have mandatory operational schemes to reduce GHG emissions in any form (e.g., cap & trade, carbon tax etc.), Projects shall only be eligible if the Project Developer has either: (a) provided Gold Standard with satisfactory justification that no double counting of emission reductions occurs or (b) has committed to retiring eligible units equal to the quantity of Gold Standard VERs. Refer to Annex A of this document.	GHG emissions in any form (e.g., cap & trade, carbon tax etc.) in Kenya, which based on section 3.1.1 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements</i> (Version 2.0) is an eligible host country. Water scarcity in various regions of Kenya has been an issue for decades. The government has issued some water resources related policies and regulations. The <i>Water Act 2016</i> provides for the regulation, management and development of water resources and water and sewerage services in line with the Constitution. The <i>National Water Quality Management Strategy (NWQMS)</i>¹⁶ issued by the Ministry of Water and Irrigation to improve water quality. The <i>Kenya National Water Master Plan 2030</i> developed by Ministry of Environment, Water and Natural Resources recognizes water as an enabler in socio-economic development and aims to improve water and sanitation availability and access to all by 2030. The implementation of the project is fully in line with the relevant water resources policies of Kenya. The project technology enables users in rural and urban regions to obtain safe drinking water through the use of a multi-layer household water filtration system without affecting the water source and supply. The implementation of this project can improve users' access to safe drinking water, which is in line with Kenya's vision of water resources. Besides, Kenya submitted its updated Nationally Determined Contribution (NDC) on 24/12/2020, which commits to abate GHG emissions by 32% by 2030 relative to the business as usual (BAU) scenario of 143 million tCO₂e¹⁷. The project is supported by the local government since it is in compliance with applicable Host Country's legal,
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¹⁶ <https://wasreb.go.ke/national-water-quality-management-strategy/>

¹⁷ [https://www4.unfccc.int/sites/ndcstaging/PublishedDocuments/Kenya%20First/Kenya's%20First%20%20NDC%20\(updated%20version\).pdf](https://www4.unfccc.int/sites/ndcstaging/PublishedDocuments/Kenya%20First/Kenya's%20First%20%20NDC%20(updated%20version).pdf)

	environmental, ecological and social regulations, benefits the local residents and will help in contributing towards Kenya's National emission reductions target. According to the document of "Letter of No Objection" signed by government of Kenya, the Project does not constitute an activity, scheme or programme that is within the scope of the Republic of Kenya's Nationally Determined Contribution dated 24 December 2020, as may be amended, updated or replaced from time to time ("NDC").
(e) Contact Details Section 3.1.1 of <i>GS4GG Principles & Requirements</i> (Version 1.2) As part of the Project Documentation the Project Developer shall provide (i) name and (ii) contact details of all Project Participants; AND in case of an organization (iii) the legal registration details and (iv) documentation by the governing jurisdiction that proves that the entity is in good standing (defined as being a legal or other appropriate entity registered in or allowed to operate within the required jurisdiction and with no evidence of insolvency or legal/criminal notices placed against it or any of its Directors). Gold Standard retains the right (at its own discretion) to refuse use of the Standard where reputational concerns are highlighted.	The Project Participants' contact details are provided in Appendix 2. Contact information of project participants of this PDD. Climate Neutral Kenya Limited is on 9 Oct 2020 Incorporated under the Companies Act, 2015 and that the Company is a PRIVATE LIMITED COMPANY (See Business license). Because the company has been established for a short time and Kenya's economic and technological level, there is no credit information on the Internet, but the company works to mitigate global climate change, mainly in the business of improving the environment. Profit Carbon Environmental Energy Technology (Shanghai) Co., Ltd. is a private owned company with limited liability instead of an organization incorporated in the Republic of China. It is in good standing as per National Enterprise Credit Information Publicity System¹⁸.
(f) Legal Ownership Section 3.1.1 of <i>GS4GG Principles & Requirements</i> (Version 1.2) Full and uncontested legal ownership of any Products that are generated under Gold Standard Certification, (for example carbon credits) shall be demonstrated. Where such ownership is transferred from project beneficiaries this must be demonstrated	Each water purifier end user, who is the default owner of emission reductions, is notified that they waive ownership of emission reductions upon sale of each water purifier. The contract will be signed with each end-user when selling the water purifier to make sure the project owner has the legal ownership of the emission reductions generated by the project activity. Refer to section A.1.2 of the PDD for further details.

¹⁸ <https://www.creditchina.gov.cn/xinyongxinxixiangqing/xyDetail.html?searchState=1&entityType=1&keyword=%E4%B8%8A%E6%B5%B7%E7%9B%88%E7%A2%B3%E7%8E%AF%E5%A2%83%E8%83%BD%E6%BA%90%E7%A7%91%E6%8A%80%E6%9C%89%E9%99%90%E5%85%AC%E5%8F%B8&uuid=f314d8d01428b2cab58f6afd8d5f3d69&tyshxydm=91310115301334442X>

transparently and with full, prior and informed consent (FPIC). Note that for certain Project types there is a requirement for full and uncontested legal land title/tenure to be demonstrated. These are contained within specific Activity or Product Requirements. All projects shall immediately report to Gold Standard any land title/tenure disputes arising. Section 3.1.4 of <i>GS4GG Community Services Activity Requirements</i> (Version 1.2) Projects involving the distribution of a large number of devices for services such as heating, cooking, lighting, electricity generation, water treatment technology such as water filter, etc. shall provide a clear description of the ownership of the Products that are generated under Gold Standard Certification all along the investment chain. In line with the FPIC requirement, the proofs that end-users are aware of and willing to give up their rights on Products shall be provided. The transfer of Product ownership shall be discussed during local stakeholder consultations for projects.	Moreover, in the stakeholder consultation meeting held on 03/09/2021, the end-users and PP discussed the ownership, and determined that the project owner had the undisputed carbon legal ownership.
(g) Other Rights Section 3.1.1 of <i>GS4GG Principles & Requirements</i> (Version 1.2) As well as legal title and ownership, the Project Developer shall also demonstrate where required uncontested legal rights and/or permissions concerning changes in use of other resources required to service the Project (for example, access rights, water rights etc.). Any known disputes or contested rights must be declared immediately to Gold Standard by the Project Developer and resolved prior to further project implementation in affected areas.	The project involves introduction of multi-Layer water purifiers as a way to provide safe drinking water for households in Makueni County, Trans Nzoia County, Kirinyaga County, Bungoma County and Nakuru County of Kenya. Households simply pour untreated water into the water purifier. Water seeps through 8 step filtration system, removing bacteria and other suspended solids, and treated water is stored in the lower tank, which is easily accessed by turning on the faucet. The water filtration system requires no energy input or consumables. The project does not involve any activity that causes alteration of any resource, or contested legal rights and other disputes, therefore the need for acquiring any specific legal right is not applicable.
(h) Official Development Assistance (ODA) Declaration	The project owner has signed the ODA declaration template and confirms that no ODA is provided under the condition that the

Section 3.1.1 of <i>GS4GG Principles & Requirements</i> (Version 1.2) All Project Developers applying for project activities located in a country named by the OECD Development Assistance Committee's ODA recipient list and seeking Gold Standard Certification for carbon credits shall declare the Official Development Assistance (ODA) support. The Project Developer shall follow the GHG Emissions Reduction & Sequestration Product Requirements and submit the declaration at the time of Design Certification. Section 6.1.1 and 6.1.2 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements</i> (Version 2.0) Projects are ineligible for carbon crediting under Gold Standard if the ODA assistance is provided to the project under the condition that the credits generated by the Project will be transferred, either directly or indirectly, to the donor country providing ODA support. Project Developer submitting a Project located in a country named by the OECD Development Assistance Committee's ODA recipient list shall sign and submit the ODA Declaration.	credits generated by the project will be transferred, either directly or indirectly, to the donor country providing ODA support.
(i) Suppressed Demand Section 3.1.3 of <i>GS4GG Community Services Activity Requirements</i> (Version 1.2) Certain Impact Quantification methodologies allow projects to account Suppressed Demand scenario when establishing a baseline. In such cases, the application of Suppressed Demand baseline is limited to Small Scale and Microscale Projects. Where a Suppressed Demand baseline is applied, it is not possible to 'stack' Gold Standard Certified Impact Statements or Products as the definition of the baseline may be contradictory.	The project activity will only claim credits for end-users that boil water and suppressed demand is not included in the baseline scenario. The same is confirmed through the baseline survey. As per Section 3.1.3 of <i>GS4GG Community Services Activity Requirements</i> (Version 1.2), it can be a large-scale GS VER project.
(j) Eligible Greenhouse Gases Section 4.1.1 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements</i> (Version 2.0) Only Carbon Dioxide (CO₂), Methane (CH₄) and/or Nitrous Oxide (N₂O) are eligible for Gold Standard crediting, provided Projects	The project only considers the reduction of Carbon Dioxide (CO₂), Methane (CH₄) and Nitrous Oxide (N₂O) Emissions for Gold Standard crediting. Refer to section B.3 of the PDD for further details.

comply with Gold Standard Requirements and eligibility criteria.	
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A.1.2 Legal ownership of products generated by the project and legal rights to alter use of resources required to service the project

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The project involves introduction of multi-Layer water purifiers as a way to provide safe drinking water for households in Makueni County, Trans Nzoia County, Kirinyaga County, Bungoma County and Nakuru County of Kenya. Households simply pour untreated water into the water purifier. Water seeps through 8 step filtration system, removing bacteria and other suspended solids, and treated water is stored in the lower tank, which is easily accessed by turning on the faucet. The water filtration system requires no energy input or consumables. The project does not involve any activity that causes alteration of any resource, or contested legal rights and other disputes, therefore the need for acquiring any specific legal right is not applicable.

The water purifier involved in this project were purchased by Climate Neutral Kenya¹⁹ Limited and sold in Kenya. So, Climate Neutral Kenya Limited has full and uncontested legal ownership of the VERs that are generated from the Gold Standard Certification of this project. Each water purifier end user, who is the default owner of emission reductions, is notified that they waive ownership of emission reductions upon sale of each water purifier. The contract signed between the project owner and the end-users will be provided as evidence.

Moreover, the end-users were informed about the transfer of carbon credits ownership through stakeholder consultation meeting, which was hold on 03/09/2021.

A.2. Location of project

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Makueni County (1°48'S and 37°37'E), Trans Nzoia County (1°6'N and 34°57'E), Kirinyaga County (0°30'S and 37°17'E), Bungoma County (0°34'S and 34°34'E) and Nakuru County (0°18'S and 36°4'E) of Republic of Kenya (Kenya has a latitude of 0.0236° S, and longitude of 37.9062° E²⁰).

¹⁹ The Climate Neutral Kenya was approved on 09/10/2020, so the signing of the Purchase framework agreement is executed by the investment company- TAL HOLDING KENYA LIMITED, and the signing date is 16/03/2020.

²⁰ <https://worldpopulationreview.com/country-locations/where-is-kenya>

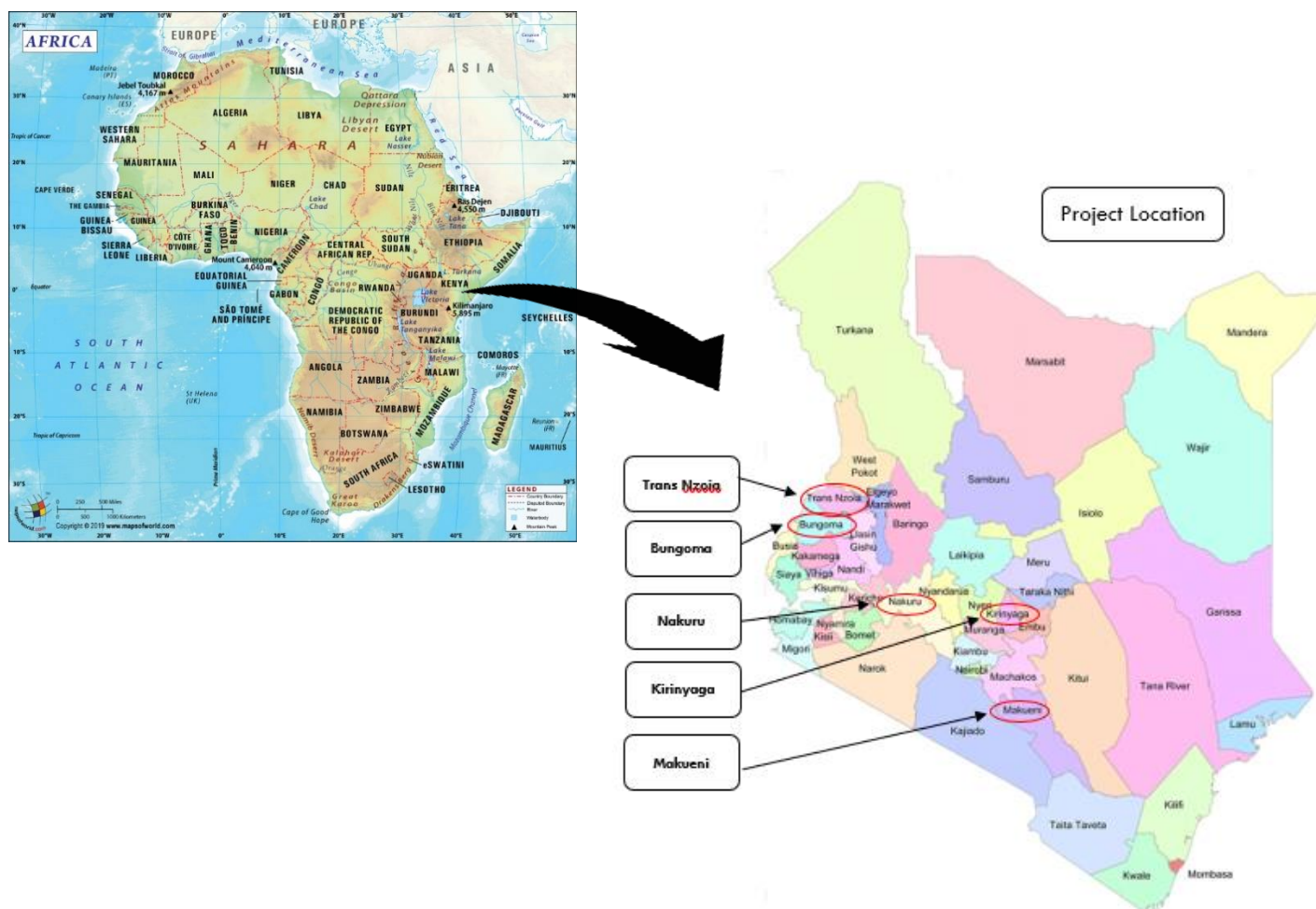


Figure A2.1 Target Area of the Project

A.3. Technologies and/or measures

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The Multi-Layer water purifiers are produced by Zhejiang Duanqi Renewable Energy Co., Ltd. which registered in China, which consist of 8 step filtration system and can purify water 3 liter/hour²¹. The sales are conducted by TRIOPT AFRICA LIMITED²² and After-sales maintenance were operated by Climate Neutral Kenya Limited. All of the materials and facilities engaged in the project are produced by same manufacture and belongs to product import. The lifespan of the water purifiers over 15 years. A whole set of the alternative dome ceramic, carbon cartridge and mineral stone will be supplied to the end-users together with the water purifiers. All the alternative filtration system can be changed easily and will not affect the usage of the water

²¹ From the "Factory inspection report", It can be confirmed that the flow rate of the water purifier involved in this project and used to emission reductions calculation is 3L/H, which Is conservative (The higher flow rate of the water filter is because of the different structure, the actual filtration surface inside the filter is larger, so the filtration efficiency is improved).

²² TRIOPT AFRICA LIMITED is a subsidiary of TAL HOLDING LIMITED (one of the funders of the project owner CNKL), so TRIOPT AFRICA LIMITED and CNKL has the same investors- TAL HOLDING LIMITED.

purifiers on that day. Considering carbon revenue of the project, all the following alternative materials needed will be provided by the project owner for free.



Figure A3.1 The using of the water purifier by local people

The project equipment provides safe drinking water that are free from pathogens and elevated levels of toxic substances. The water treated by the multi-layer water purifiers are compliance with the local standard of *KS EAS 153: 2018 East African Standard Specification for Packaged Drinking Water*. All water directly intended for drinking are not have detectable E.coli in any 100 ml sample, which will also comply with WHO Guideline values for verification of microbial quality²³.

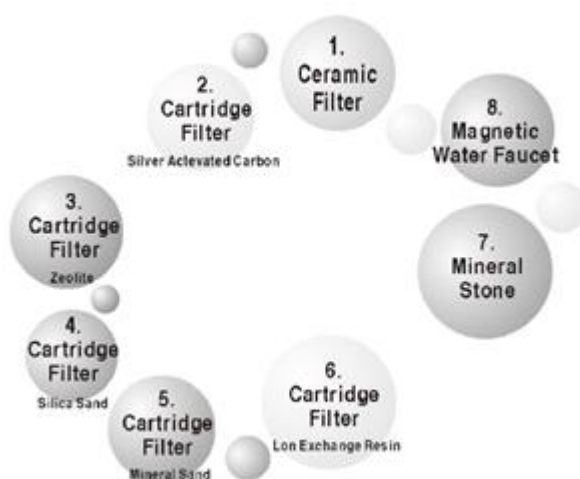


Figure A3.2 8 Step Filtration System

²³ Table 7.10, page 149. World Health Organization. (2017). Guidelines for Drinking-water Quality: fourth edition incorporating the first addendum. Geneva: World Health Organization.

Dome ceramic filter

The ceramic filter is made of highly compressed ceramic diatomic substances. It has a very tiny porous structure of 0.2µm, which is less than half size of bacterias (0.5-1.0µm). Only with this filter, it will get water free from rust, dust, sand and bacterias such as cholera, typhoid, coliform etc.

Granular activated carbon

Most of the home drinking water filters in the market are granular activated carbon filters. Carbon has a long history of being used to absorb impurities and is the most powerful adsorbent known. One pound of carbon has a square area of 125 acres and can absorb thousands of different chemicals.

Zeolite

Zeolite's are natural volcanic material with a number of unique characteristics. Zeolite's have an unusual crystalline structure and a unique ability to exchange cations which are positively charged ions. This substitution of ions enables zeolite to selectively absorb certain harmful and unwanted element from water. Zeolite exchange sodium ions for calcium ions which results in soft water. Zeolite also have a strong affinity for harmful heavy metals such as mercury, lead and chromium.

Silica sand

Silica is the mineral building block that promotes healthy bone and connective tissue and helps prevent premature aging and cardiovascular and Alzheimer's disease. It's porous structure also enables adsorption of bad elements in water. And in one of its unique characters, it always keeps temperature of itself so as to keep the water cool and fresh.

Mineral Sand

Mineral is the general name of various elements necessary to constitute human tissues and maintain normal physiological functions. Mineral sand is used to regulate the content of inorganic minerals in water.

Ion Exchange Resin

Ion exchange resin is a kind of polymer with corresponding functional groups. In general, conventional sodium ion exchange resins contain a large amount of sodium ions. When the content of calcium and magnesium ions in water is high, the ion exchange resin can release sodium ions, and the functional groups combine with calcium and magnesium ions, so that the

content of calcium and magnesium ions in water is reduced and the hardness of water is reduced. Hard water becomes soft water. so, it is mainly used to adjust the hardness of water.

Mineral stone

The mineral stone has been regarded as a mysterious stone that has a plenty of mineral supplement. It has been widely used in health-related products for centuries. It is composed of many health effective elements such as potassium, sodium, calcium and magnesium. It will keep the water stay fresh with full of mineral.

Magnetic Water Tap

The earth's magnetic field generally promotes vitality for the human body. To take advantage of this, the outlet of drinking water system is fixed with magnets. The magnetic power changes the water to have the free moving hexagonal structured water molecules in harmony with fluids in human body.

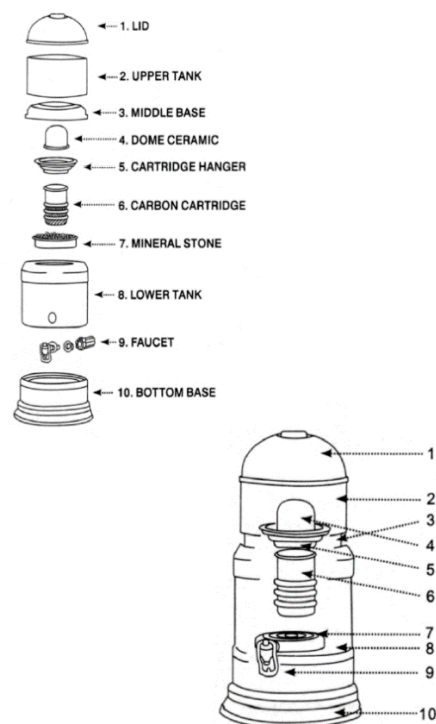


Figure A3.3 Components of the Water Purifier

The project activity will contribute mainly towards the SDGs in the following aspects:

SDG 6 Clean Water and Sanitation

The water purifiers of these projects will offer an affordable, long-term and zero emission solution for households that generally consume unsafe drinking water. With a capacity to purify

3 liters of water per hour over a 15-year lifespan of the water purifiers, the project technology can provide enough water to meet daily water consumption of the local residents.

SDG 7 Affordable and Clean Energy

The project involves introduction of multi-Layer water purifiers as a way to provide safe drinking water for households in Makueni County, Trans Nzoia County, Kirinyaga County, Bungoma County and Nakuru County of Kenya. Air pollution caused by combustion of woody fuels (baseline scenario) will be reduced since there will be no more boiling of water (project scenario). Using the purifiers will reduce the diseases associated with air pollution or water borne.

SDG 8 Decent Work and Economic Growth

The project activity will provide job opportunities for all local residents during project implementation (dissemination and maintenance of the purifiers) and monitoring activities irrespective of gender or any other status.

SDG 13 Climate Action

The project will not only dramatically increase access to safe drinking water but also reduce consumption for woody fuels previously required to treat drinking water, which will decrease environmental degradation and greenhouse gas emissions. Besides, the project will provide an opportunity for local residents to learn and raise awareness on climate change, mitigation measures and how to reduce impacts such as combating deforestation.

A.4. Scale of the project

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As per section A1.1 of this PDD, the annual average thermal energy savings is 1,056.042 GWh_{th}, which is more than 180 GWh_{th}, as per section 9.1.1 and 9.1.2 of *GS4GG GHG Emissions Reduction & Sequestration Product Requirements (Version 2.0)*, it's a large-scale GS VER project.

A.5. Funding sources of project

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The project will be financed by the project owner (Climate Neutral Kenya Limited) only during the entire operational lifetime. No public funding will be involved. On the long-run the project will be financially sustainable through the sale of water purifiers and carbon credits.

Project financing for this project activity will not use Official Development Assistance (ODA) Funds. The signed ODA declaration template will be provided as evidence.

SECTION B. APPLICATION OF APPROVED GOLD STANDARD METHODOLOGY (IES) AND/OR DEMONSTRATION OF SDG CONTRIBUTIONS

B.1. Reference of approved methodology (ies)

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Approved methodology applied: *Methodology for Emission Reductions from Safe Drinking Water Supply* (Version 1.0).

<https://globalgoals.goldstandard.org/429-ee-sws-emission-reductions-from-safe-drinking-water-supply/>

B.2. Applicability of methodology (ies)

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Justification for the choice of the selected methodology is shown in the following table:

Methodology for Emission Reductions from Safe Drinking Water Supply (Version 1.0)	
Eligibility Criteria	Justification for the Project Activity
This methodology is applicable to project activities that introduce a new, or rehabilitate an existing, zero-emission or low-emission technology to supply safe drinking water.	The project will introduce zero-energy water filtration technology which will displace the use of woody fuels traditionally used to treat drinking water. The water filtration system requires no energy input or consumables. It's a kind of gravity household water purifiers. The project technology is applicable.
Technologies include household water treatment technologies (HWT), Institutional water treatment technologies (IWT), Community level water treatment technologies (CWT) and community water supply technologies (CWS). The methodology provides two sets of calculation methods and monitoring requirements, one set that applies to the HWT and IWT types of technologies, and another set that applies to the CWT and CWS types of technologies.	The project involves introduction of household water purifiers, which belong to household water treatment technologies (HWT).
Under this Methodology, a project's objectives are to reduce or avoid greenhouse gas emissions from boiling unsafe drinking water in the baseline, and to supply drinking water that is	The project activity will introduce safe water supply technology for households in Makueni County, Trans Nzoia County, Kirinyaga County, Bungoma County and

safe for consumption when it enters the project households or institutional premises. When the drinking water is treated in the household or institution (HWT or IWT), then the water supplied from the treatment technology should be safe. When the water is supplied or retrieved from a CWT or CWS directly to the premises of the household or institution, then the water entering the end-user premises should be safe.	Nakuru County of Kenya. The water purifiers of the project will offer an affordable, long-term and zero emission solution for households that generally consume unsafe drinking water. It will not only dramatically increase access to safe drinking water but also reduce consumption for woody fuels previously required to treat drinking water, which will decrease environmental degradation and greenhouse gas emissions. Water quality test will be conducted annually by accredited laboratory based on <i>KS EAS 153: 2018 East African Standard Specification for Packaged Drinking Water</i>. The sample size will follow 90/10 precision rule and the minimum size is 30. The water samples will be randomly selected from the project database and taken from outlet of the water purifiers.
Eligible household water treatment technologies (HWT), institutional water treatment technologies (IWT), and community level water treatment technologies (CWT) include bleach/chlorine, water filter (ceramic, sand, composite, membrane, etc.), UV disinfection, etc.	The project introduces zero-energy water filtration technology for local residents, which belongs to household water treatment technologies (HWT). The water filtration system requires no energy input or consumables. It's a kind of gravity household water purifiers. The project technology is applicable.
Eligible community water supply technologies (CWS) include new installation of new borehole hand-pumps, borehole hand-pumps rehabilitation, solar powered drinking water pumps, etc. Water pumps powered by fossil-fuel engines are not eligible, with the exception of backup fossil-fuel engines that are used for no more than 10% of operating hours (parameter SWDS 33).	No community water supply technologies (CWS) will be involved in the project activity, this condition is not applicable.

All projects involving CWT and CWS technologies must also include ongoing maintenance and repair of the project technology.	Neither community level water treatment technologies (CWT) nor community water supply technologies (CWS) will be involved in the project activity, this condition is not applicable.
Where the project involves the rehabilitation of an existing technology, the project developer shall provide evidence that the existing technology is non-operational and that there is no planned maintenance or repair for at least 3 months after the date it became non-operational (parameter SWDS 2).	No rehabilitation of an existing technology will be involved in the project activity, this condition is not applicable.
This methodology allows for project activities to include safe water treatment and/or supply technologies implemented for end-users in households, and/or commercial premises such as shops or institutional premises including half or full day/boarding schools, prisons, army camps & refugee camps.	The project activity will introduce zero-energy safe water treatment technology for end-users in households.
In cases where the safe water is retrieved at the CWT or CWS location, the water in its improved form shall be available within a distance of 1 km or less from the end-users, as demonstrated by satellite imaging or GPS coordinates of each CWT or CWS location. Alternatively, as a proxy, a total collection time of 30 minutes or less for a round trip, including queuing, using the travel modes of walking or pedaling may be demonstrated (parameter SDWS 1).	Neither community level water treatment technologies (CWT) nor community water supply technologies (CWS) will be involved in the project activity, this condition is not applicable.
Project technology performance level (HWT and IWT): It shall be demonstrated based on report of laboratory testing or official notification that the project technology or equipment achieves either (i) the performance target classification 3-star or 2-star level, meaning "Comprehensive	The water provided by the project activity will conform to drinking water quality specified in relevant local microbiological quality standards and WHO Guideline values for verification of microbial quality ²⁴ . The water purifier involved in this

²⁴ All water directly intended for drinking must not have detectable E.coli in any 100 ml sample. Table 7.10, page 149 of *Guidelines for Drinking-water Quality: fourth edition incorporating the first addendum*. World Health Organization 2017. <https://www.who.int/publications/i/item/9789241549950>.

Protection,” as per the WHO International Scheme to Evaluate Household Water Treatment Technologies⁸ (World Health Organization, 2011) or (ii) compliance with the national standard or guideline for household drinking water treatment technology; if no national guideline or standard is available, then the project technology shall comply with the WHO International Scheme requirements as per (i) (parameter SDWS 2).	project obtain the pre-export verification of conformity by KEBS (Kenya bureau of standards). The pre-export verification of conformity by KEBS shows that the water purifier products are allowed to be introduced into Kenya. Moreover, according to the water quality test report by the KEBS, the purified water quality of the product meets the standards of KS EAS 153: 2018 East African Standard Specification for Packaged Drinking Water and the Guidelines for Drinking-water Quality: fourth edition incorporating the first addendum. World Health Organization 2017. The sample size will follow 90/10 precision rule and the minimum size is 30. The water samples will be randomly selected from the project database and taken from outlet of the water purifiers.
Project technology performance level (CWT and CWS): For each individual CWT or CWS, it shall be demonstrated at the start of each crediting period with water quality testing reports that the water directly supplied by the project water technology/source achieves both: microbial quality in line with either (i) national standards or guidelines for microbial quality of drinking water, or in the absence of such requirements, (ii) the guideline values for verification of microbial quality from the Guidelines for drinking-water quality (Table 7.10, WHO, 2017); and compliance with (i) national standards or guidelines on priority chemical contamination and physical and aesthetic aspects, or in the absence of such requirements, (ii) international standards or guidelines on priority chemical	Neither community level water treatment technologies (CWT) nor community water supply technologies (CWS) will be involved in the project activity, this condition is not applicable.

contamination and physical and aesthetic aspects. (Parameter SWDS 3).	
The project must conduct annual water hygiene education campaigns for the end-users. (Parameter SDWS 20).	Hygiene education campaigns will be carried out annually. Refer to section B.7.3 of this PDD.
A project applying this methodology may make SDG claims if relevant monitoring parameter(s) is included in the monitoring plan to demonstrate and confirm the project's contributions to SDGs. See parameter SDWS 19.	Relevant monitoring parameters for SDG 6, SDG7, SDG 8 and SDG 13 are included in the monitoring plan. Refer to section B.7 of the PDD for more details.

B.3. Project boundary

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The project boundary is specified in section 3.1 of the applied GS methodology as follows:

The project boundary includes:

- the physical, geographical sites of the low- or zero-greenhouse gas emitting technologies to treat/supply safe drinking water installed by the project activity,
- any back-up engines or other equipment using fossil-fuel related to the low greenhouse gas emitting technologies,
- the electricity grid, in the case electricity is used by the project, and
- the household, commercial and institutional buildings where the end users of safe water provided by the project are located.

The project will introduce zero-energy water filtration technology which will displace the use of woody fuels traditionally used to treat drinking water. The water filtration system requires no energy input or consumables. The project boundary is therefore defined as the location of the water purifiers and the environs within which purified water from the project technology are consumed. The target area for this project is defined by the regions in which the project activity is to be implemented, which is Makueni County, Trans Nzoia County, Kirinyaga County, Bungoma County and Nakuru County of the Republic of Kenya. Refer to section A.2 of the PDD for more details. The greenhouse gases included in or excluded from the project boundary are shown in Table B3.1 below.

Table B3.1 Emission Sources Included in or Excluded from the Project Boundary

Source	GHGs	Included?	Justification/Explanation
Baseline	CO ₂	Yes	Major source of emissions
	CH ₄	Yes	Minor source of emissions

Project scenario	Emissions from wood fuels utilized for obtaining safe drinking water displaced due to project activity	N ₂ O	Yes	Minor source of emissions
	Emissions from fossil fuels utilized for obtaining safe drinking water displaced due to project activity	CO ₂	Yes	Major source of emissions
		CH ₄	No	Excluded for simplification
		N ₂ O	No	Excluded for simplification
	Emissions from electricity for operating project water supply/treatment technology	CO ₂	No	No electricity will be required in the project scenario
		CH ₄	No	No electricity will be required in the project scenario
		N ₂ O	No	No electricity will be required in the project scenario
	Emissions from fossil fuels for operating project water supply/treatment technology	CO ₂	No	No fossil fuels will be required in the project scenario
		CH ₄	No	No fossil fuels will be required in the project scenario
		N ₂ O	No	No fossil fuels will be required in the project scenario

B.4. Establishment and description of baseline scenario

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As per section 3.4 of the applied GS methodology, the general baseline scenario is that users would have boiled drinking water in the absence of the project activity. The project activity will only claim credits for end-users that boil water and suppressed demand is not included in the baseline scenario. As per section 3.5.1 of the applied GS methodology, the applicable baseline scenario for fuel and technology is determined based on section 2 Selection of baseline scenarios and project scenarios of *Technologies and Practices to Displace Decentralized Thermal Energy Consumption* (TPDDTEC, Version 3.1).

B.4.1 Baseline survey (BS) of target population characteristics

The baseline survey provides critical information on the following pre-project conditions:

1. Practices of boiling water or drinking unsafe water: Document the safe drinking water sources and/or treatment technologies available and used in the project boundary (e.g. no treatment, boiling, piped water, chlorine tablets, etc.). This critical information corresponds to the Percentage of households available of safe drinking water in baseline survey.

2. Efficiency of water boiling systems: Document the stove or water boiling technologies used in the project boundary. This critical information corresponds to the Percentage of households treat unsafe water by boiling and the Efficiency of water boiling systems.
3. Baseline fuels: Document the cooking fuels used in the project boundary. This critical information corresponds to the Percentage of households treat unsafe water by boiling and the Percentage of fuel type for each type stove.

The baseline survey requires in person interviews with a robust sample of local households in the target project area. As per *TPDDTEC* (Version 3.1), the baseline survey should be carried out for each baseline scenario using representative and random sampling, following these guidelines for minimum sample size:

Group size <300: Minimum sample size 30 or population size, whichever is smaller

Group size 300 to 1,000: Minimum sample size 10% of group size

Group size > 1,000 Minimum sample size 100

As per *2019 Kenya Population and Housing Census Volume I: Population by County and Sub-County*²⁵, There are 244,669 households in Makueni County, 223,808 households in Trans Nzoia County, 204,188 households in Kirinyaga County, 358,796 households in Bungoma County and 616,046 households in Nakuru County of Kenya. Therefore, the sample size should be more than 100.

For baseline survey, 90/10 precision level was selected. Calculation of the sample size for the survey was based upon a 0.6 Coefficient of Variation (COV) under a 90/10 approach as provided in Table 3 of Annex 4 of *TPDDTEC* (Version 3.1). Table 3 is applicable because the survey was conducted for the baseline scenario as a single sample. The selection of a COV of 0.6 was arbitrarily determined from the project proponent's assumption which is allowed in Annex 4 of *TPDDTEC* (Version 3.1) "Use the tables here to choose a provisional minimum sample size using that COV estimate." In summary, 101 households were randomly sampled across each County of the target project area, 505 households were surveyed totally.

The survey was carried out from 10/10/2020 to 20/12/2020 by the project owner. The sampling questionnaire used in the interview is shown as the following table.

Table B4.1 Baseline Survey Questionnaire

Name		Address		Telephone Number	
1. How many individuals are there in the household?					

²⁵ <https://www.knbs.or.ke/?wpdmpo=2019-kenya-population-and-housing-census-volume-i-population-by-county-and-sub-county>

2. How do you usually collect drinking water? (piped water supply, boreholes, shallow wells, river, rain water and etc.)
3. What's the technologies used daily to treat drinking water? (available of safe water, non-GHG emitting technologies like chlorine treatment techniques, using water filter, boiling water, drinking unsafe water directly and etc.)
4. What's the water treatment technology usage patterns and tasks? (commercial, institutional, domestic and etc.)
5. If you can get safe water (treated or from safe supply), will you still boil it before drinking?
6. If you drink boiled water, what fuels does your family usually use?
7. What's the sources of fuels? (purchased, collected from the forest and etc.)
8. If the fuels are collected from the forest, how far is the walking distance?
9. What kind of stoves does your family usually use to boil water? (Three-stone stoves, conventional stoves, improved stoves, LPG stoves and etc.) If LPG stove is used, please provide model of it.
10. Whether the seasonal variations will affect the technologies chose and fuel used to treat drinking water?

Average household size

A total of 505 households participated in the baseline survey, which included 1,994 individuals. The number of individuals per household can be calculated as $1,994/505=3.9$ individuals/household. This parameter is used to the ex-ante calculation of Volume of drinking water per person per day (QPW_p) which the parameter ID is SDWS 24 in section B.7.1.

Water sources and amount for drinking

0.59% (3HH/505HH) of the households are available of piped water. 6.7% (34HH/505HH) of the households take water from boreholes or protected wells. As per Annex 2 of the applied GS methodology, piped water and water from boreholes or protected wells belong to improved sources of drinking water. For the rest of 92.7% (468HH/505HH) households, the most common sources of water are from unprotected well, unprotected spring and surface water, which belong to unimproved sources of drinking water. This parameter is corresponding to the *Water sources in the project boundary* which the parameter ID is SDWS 5 in section B.6.2.

According to *Water Survey Report* approved by National Environment Management Authority in Turkana, Makueni, Kirinyaga, Mandera, Tran Nzoia, Bungoma and Nakuru County, the local residents typically need 16.8 liters of water for drinking per household per day. With an average of 3.9 person per household, the number of liters needed for drinking per person per day can be calculated as $16.8/3.9=4.3$ L/person/day.

Practices of boiling water or drinking unsafe water

The percentage of households who are available of safe drinking water before project activity is 7.3% (37HH/505HH). 78.4% (396HH/505HH) treat unsafe water before drinking and 14.3% (72HH/505HH) drink unsafe water directly without any treatment. Data and information collected from the questionnaires show that all the households surveyed are involved in water treatment for domestic consumption. Among those who treat water, 83.8% (332HH/396HH) of them boil water before drinking and 16.2% (64HH/396HH) use gravity household water filters, ultraviolet radiation treatment, reverse osmosis, chlorine tablets and other technologies. Therefore, C_b (proportion of project households who in the baseline were already using a safe water supply that did not require boiling it) can be calculated as $7.3\% + 78.4\% \times 16.2\% = 20.0\%$. None of the households (0%) will still boil water before drinking if they can get safe water (treated or from safe supply).

Efficiency of water boiling systems and baseline fuels

Among those who treat drinking water by boiling, the main fuel used are non-renewable biomass of firewood (agriculture waste is included, 75.9%, 252HH/332HH), charcoal (12.3%, 56HH/332HH) and LPG²⁶ (7.2%, 24HH/332HH). 55.2% (139HH/252HH) of firewood are collected from the nearby woodland by women and children. 56.1% (78HH/139HH) of them spend over 30 minutes per day to collect firewood. The stove used to boil water in baseline scenario is three-stone stoves (67.2%, 223HH/332HH), Conventional stoves (19.3%, 64HH/332HH), improved stoves (6.3%, 21HH/332HH) and LPG stoves (7.2%, 24HH/332HH). 85.1% (430HH/505HH) of the households consider that the seasonal variations will not affect the technologies chose and fuel used treat drinking water. Breakdown of fuel type against type of stove of each County are shown as the following tables.

Table B4.2 Breakdown of Fuel Type against Type of Stove-Makueni County

Fuel Type	Type of Stove-Makueni County				
	Three-stone stoves	Conventional stoves	Improved stoves	LPG stoves	Total
Firewood	75.0% (36HH/48HH)	4.2% (2HH/48HH)	2.1% (1HH/48HH)	0%	81.3%
Charcoal	0%	16.7% (8HH/48HH)	2.1% (1HH/48HH)	0%	18.8%
LPG	0%	0%	0%	0%	0%
Total	75.0%	20.8%	4.2%	0%	100%

²⁶ Liquefied petroleum gas.

Table B4.3 Breakdown of Fuel Type against Type of Stove-Trans Nzoia County

Fuel Type	Type of Stove-Trans Nzoia County				
	Three-stone stoves	Conventional stoves	Improved stoves	LPG stoves	Total
Firewood	65.2% (30HH/46HH)	10.9% (5HH/46HH)	2.2% (1HH/46HH)	0%	78.3%
Charcoal	0%	15.2% (7HH/46HH)	4.3% (2HH/46HH)	0%	19.6%
LPG	0%	0%	0%	2.2% (1HH/46HH)	2.2%
Total	65.2%	26.1%	6.5%	2.2%	100%

Table B4.4 Breakdown of Fuel Type against Type of Stove-Kirinyaga County

Fuel Type	Type of Stove-Kirinyaga County				
	Three-stone stoves	Conventional stoves	Improved stoves	LPG stoves	Total
Firewood	88.9% (64HH/72HH)	4.2% (3HH/72HH)	0%	0%	93.1%
Charcoal	0%	5.6% (4HH/72HH)	1.4% (1HH/72HH)	0%	6.9%
LPG	0%	0%	0%	0%	0%
Total	88.9%	9.7%	1.4%	0%	100%

Table B4.5 Breakdown of Fuel Type against Type of Stove-Bungoma County

Fuel Type	Type of Stove-Bungoma County				
	Three-stone stoves	Conventional stoves	Improved stoves	LPG stoves	Total
Firewood	69.6% (48HH/69HH)	7.2% (5HH/69HH)	2.9% (2HH/69HH)	0%	79.7%
Charcoal	0%	13.0% (9HH/69HH)	4.3% (3HH/69HH)	0%	17.4%
LPG	0%	0%	0%	2.9% (2HH/69HH)	2.9%
Total	69.6%	20.3%	7.2%	2.9%	100%

Table B4.6 Breakdown of Fuel Type against Type of Stove-Nakuru County

Fuel Type	Type of Stove-Nakuru County				
	Three-stone stoves	Conventional stoves	Improved stoves	LPG stoves	Total
Firewood	46.4% (45HH/97HH)	7.2% (7HH/97HH)	3.1% (3HH/97HH)	0%	56.7%
Charcoal	0%	14.4% (14HH/97HH)	7.2% (7HH/97HH)	0%	21.6%
LPG	0%	0%	0%	21.6% (21HH/97HH)	21.6%
Total	46.4%	21.6%	10.3%	21.6%	100%

B.4.2 Baseline non-renewable biomass (NRB) assessment

Biomass fuels are the largest source of primary energy in Kenya with woody fuels accounting for about 69% of the total primary energy consumption. About 55% of this is derived from farmlands in the form of woody biomass as well as crop residue and animal waste and the remaining 45% is derived from forests²⁷. The current forest cover of the land area in Kenya is 7.4%, which is still below the constitutional requirement of 10%. Kenya's annual population growth stands at 2.3%. Increasing population has brought immense pressure to bear on forests especially in high potential areas²⁸.

As firewood and charcoal are baseline fuels, the fractional non-renewability of biomass needs to be assessed. As per section 4 of applied methodology, fractional non-renewability status of woody biomass fuels (f_{NRB}) can be determined by CDM Tool 30: calculation of the fraction of non-renewable biomass (Version 03.0).

The specific procedure is as follows:

(1) Project participants and DNAs shall identify and clearly delineate the applicable area for which f_{NRB} is determined. For project- or PoA-specific f_{NRB} values, project participants shall assess and use the area from which woody biomass is sourced for end-users included in the project activity and justify the selection of the area in CDM project design documents. For region- or country-specific f_{NRB} values, DNAs shall specify the applicable area.

The proposed project is implemented in five counties in Kenya, since it is difficult to estimate the locations and extent of areas from which woody biomass fuel used by the project is collected

²⁷ National Energy Policy, Ministry of Energy, Republic of Kenya, October 2018. https://kplc.co.ke/img/full/BL4PdOqKtxFT_National%20Energy%20Policy%20October%20%202018.pdf

²⁸ Draft National Forest Policy 2020, Ministry of Environment and Forest, Republic of Kenya, May 2020. <http://www.environment.go.ke/wp-content/uploads/2020/06/Draft-Forest-Policy-19May-2020-.pdf>

and the Data on wood consumption at the county level are not available, therefore, data at the national level in Kenya are used for the determined of f_{NRB} .

(2) Project participants and DNAs may choose between the following two options to determine the value of f_{NRB} :

(a) Use a default value of 0.3; or

(b) Calculate f_{NRB} by determining the share of renewable and non-renewable woody biomass in the total quantity of woody biomass consumption for the country/region or the project area (hereinafter referred as the applicable area) following the procedure and requirements in the paragraphs below.

PP selected the option (b) to determine the value of f_{NRB} .

(3) As per paragraph 7 of the CDM Tool 30, the fraction of woody biomass that can be estimated as follows:

$$f_{NRB} = \frac{NRB}{(NRB + RB)} \quad (1)$$

Where:

f_{NRB}	Fractional of non-renewable biomass in the applicable area in the relevant period (fraction or %)
RB	Quantity of renewable biomass that is available on a sustainable basis in the applicable area in the relevant period (tonnes)
NRB	Quantity of non-renewable biomass consumed in the applicable area in the relevant period (tonnes)

According to step (7) below, the RB is 243PJ, NRB is 724PJ-243PJ=481 PJ, so the $f_{NRB} = 481/(481+243)=66.4\%$

(4) The relevant period should be one year, unless the methodology referring to this tool specifies otherwise.

The data used in the project calculation are annual data.

(5) The value of f_{NRB} for the applicable area shall be calculated using either of the two following options: (a) Ex ante: the f_{NRB} value is determined once at the validation stage, thus no monitoring and recalculation of the f_{NRB} value during the crediting period is required; (b) Ex post: the $f_{NRB,y}$ value is determined for the year y in the crediting period, requiring the f_{NRB} value to be updated annually, following a consistent calculation procedure throughout the crediting period.

The Ex ante will be selected by PP.

(6) In the case of ex ante calculation of f_{NRB} , the parameter f_{NRB} shall be estimated using the most recent historical year for which data is available. In the case of ex-post calculation of f_{NRB} , the parameter f_{NRB} shall be estimated for the applicable year y of the crediting period. Where available, the same vintage of data should be used for all parameters applied in this tool to calculate f_{NRB} . Where data for one single vintage is not available for all parameters, different vintages may be used for parameters, as long as it can be justified (e.g., the use of different vintages leads to a conservative estimate of f_{NRB}).

In the case of ex ante calculation of f_{NRB} , the parameter f_{NRB} will be estimated using the most recent historical year for which data is available. The project proponent aggregates all reachable woody biomass fuel collection areas within Kenya and apply a single fraction derived from all collection areas in the country.

(7) The quantity of non-renewable biomass consumed in the applicable area (NRB) shall be determined as the difference between the total consumption of woody biomass in the applicable area (H) and the quantity of renewable biomass that can be sustainably harvested in the applicable area (RB):

$$NRB = H - RB$$

Where:

H Total consumption of woody biomass in the applicable area in the relevant period (tonnes)

According to *East African Civil Society for Sustainable Energy and Climate Action* (EASE & CA) published on August 2020²⁹, biomass fuels consumption in Kenya in 2017 was 724 PJ³⁰, in which sustainable potential of solid biomass was 243 PJ. i.e., H is 724 PJ, RB is 243PJ, so the NRB is $724PJ - 243PJ = 481$ PJ.

(8) Cubic meters (m^3) instead of tonnes (t) of wood may be used for estimation of f_{NRB} values, as long as the same unit and the same conversion factor (e.g., wood density, moisture content) are consistently used for estimation of both the total consumption of woody biomass (H) and the renewable biomass available (RB).

The same unit will be used for estimation of both the total consumption of woody biomass (H) and the renewable biomass available (RB). For this project, the applied unit of the total consumption of woody biomass (H) and the renewable biomass available (RB) is PJ.

(9) If the f_{NRB} value is estimated at the national level, as a cross check, project proponent shall

²⁹ https://www.inforse.org/africa/pdfs/Pub_100-Renewable-Energy-Plan-for-Kenya-by-2050-12-08-2020.pdf

³⁰ $1PJ = 10^{15}J$.

compare the value of estimated NRB with the product of: i) total average above ground biomass tonnage of the area of forest areas deforested in recent past (tonnes/ha), and ii) most recent available observed annual rate of deforestation (ha/yr). If the estimated NRB value is more than 10% above the value calculated as per the product of biomass and deforestation rate, justification shall be provided for the higher value for NRB.

As the total average above ground biomass tonnage of the area of forest areas deforested in recent past can't be obtained from the opened document on internet, so the cross check applied this method can't be applied. While, according to the "default country-specific f_{NRB} values approved by the Board which have already expired (<https://cdm.unfccc.int/DNA/fNRB/index.html>)", the f_{NRB} in Kenya is 92%. The f_{NRB} used in the project of GS1078 is also 92%, in the proposed project, the f_{NRB} is 66.4%, which is conservative.

In conclusion, the f_{NRB} of 66.4% is fixed based on the results of the NRB assessment for the first crediting period.

B.5. Demonstration of additionality

As per section 3.6.2 of the applied GS methodology, the amount of energy required to boil 1 L of water is 360.83 kJ/L. According to equation 7 and 8 of section B.6 of the PDD, The annual quantity of safe water provided by each of the project water purifier (solely composed of isolated units of the project activity) can be calculated as $U_{p,y}$ (Usage rate of the project technology by premises type p during year y) * $QPW_{hh,p,y}$ (Volume of drinking water per premises p per day in year y) * $DP_{p,y}$ (Days the project technology is present for end-users in the premises p in year y) = $80\% * 15 * 365 = 4,380$ L. Based on the results of baseline survey, the weighted average efficiency of the baseline stove for water boiling is 16.1%. Annual thermal energy savings can be calculated as $360.83 \text{ kJ/L} * 4,380 \text{ L} / 1,000,000 / 16.1\% = 9.816 \text{ GJ} = 8.180 \text{ MWh}_{th}$ ³¹, which is less than 1,800 MWh_{th} (600 MWh). Besides, End-users of the water purifiers are households. The project activity falls under section 1.1.3 of Annex B Positive list of *GS4GG Community Services Activity Requirements (Version 1.2)*. As per section 4.1.9(a) of *GS4GG Community Services Activity Requirements (Version 1.2)*, the project considered as deemed additional.

B.5.1 Prior Consideration

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According to Gold Standard's Principles and Requirements (version 1.2), Retroactive projects shall submit the required documents for preliminary review (time of first submission) within one year of the project start date. Retroactive Project submitted at a date later than one year from the project start date will not be eligible for Gold Standard certification.

³¹ Unit conversion from GJ to $\text{MWh}_{th} = 3/3.6 = 0.833$, i.e., $1 \text{ GJ} = 0.833 \text{ MWh}_{th}$.

The project start date is 09/10/2020, and PP have submitted the required documents for preliminary review on 27/05/2021 which within one year of the project start date. Therefore, the project is eligible for Gold Standard certification.

The main action made for the project implementation are generalized in the following:

Table B5.1 Milestone of the project

Time	Milestone
07/08/2020	Completion of the project proposal (Including sales plan for the water purifiers and prior consideration of carbon revenue).
25/08/2020	Conducted business management meeting determining to apply for Gold Standard certification of this project.
09/10/2020	Start date of the project (first purchase receipt for the water purifier sold to an end-user).
27/05/2021	The first submission date to GS
03/09/2021	Stakeholder consultation physical meeting

B.5.2 Ongoing Financial Need

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The project is neither under Design Certification Renewal nor required to demonstrate financial additionality. Ongoing financial need is not applicable.

B.6. Sustainable Development Goals (SDG) outcomes

Relevant Target/Indicator for each of the four SDGs

Sustainable Development Goals Targeted	Most relevant SDG Target	SDG Impact Indicator (Proposed or SDG Indicator)
6 Clean Water and Sanitation	Target 6.1 By 2030, achieve universal and equitable access to safe and affordable drinking water for all With a capacity to purify 3 liters of water per hour over a 15-year lifespan of the water purifiers, the project technology can provide enough water to meet daily water consumption of the local residents.	6.1.1 Proportion of population using safely managed drinking water services Project Monitoring Indicator: Total number of person.days served with satisfactory level of safe water.
7 Affordable and Clean Energy	7.2 By 2030, increase substantially the share of renewable energy in the global energy mix Air pollution caused by combustion of woody fuels (baseline scenario) will be	7.2.1 Renewable energy share in the total final energy consumption. Project Monitoring Indicator: The indicators of perceived smoke levels, incidence of coughing and incidence

reduced since there will be no more boiling of water (project scenario). Using the purifiers will reduce the diseases associated with air pollution or water borne.

of itchy eyes would have varied responses from different end-users. Accordingly monitoring of this indicator would be rendered impossible to standardize. The number of water purifiers sold each year will be monitored.

8 Decent Work and Economic Growth	Target 8.5 By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value The project activity will provide job opportunities for all locals during project implementation (dissemination and maintenance of the purifiers) and monitoring activities irrespective of gender or any other status. Equal pay for work of equal value will be made to both men and women.	8.5.2 Unemployment rate, by sex, age and persons with disabilities Project Monitoring Indicator: 1) Number of males and females employed by the project. 2) Percentage of employees with salaries paid at par with the average wage of Kenya.
13 Climate Action	Target 13.3 Improve education, awareness-raising and human and institutional capacity on climate change mitigation, adaptation, impact reduction and early warning The project will not only dramatically increase access to safe drinking water but also reduce consumption for woody fuels previously required to treat drinking water, which will decrease environmental degradation and greenhouse gas emissions. Besides, the project will provide an opportunity for local residents to learn and raise awareness on climate change, mitigation measures and how to reduce impacts such as combating deforestation.	13.3.2 Number of countries that have communicated the strengthening of institutional, systemic and individual capacity-building to implement adaptation, mitigation and technology transfer, and development actions Project Monitoring Indicator: GHG emission reductions

B.6.1 Explanation of methodological choices/approaches for estimating the SDG Impact

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SDG contribution and project monitoring indicator	Project monitoring indicator	Baseline outcome	Project outcome
SDG 6 Clean Water and Sanatation	Total number of person.days served with satisfactory	In baseline scenario, no project water purifiers will be	In the project scenario, total number of person.days served with satisfactory level of safe water can be calculated as follows:

	level of safe water.	used. Therefore, the baseline outcome benefit is 0.	Project outcome of SDG 6 = $N_{p,y} * HN_{p,y} * DP_{p,y} * U_{p,y} * M_{q,y}$ (1) Where: <table><tr><th>Parameters</th><th>Description</th><th>Source/Value</th></tr><tr><td>Project outcome of SDG 6</td><td>Total number of person.days served with satisfactory level of safe water</td><td>Calculated based on equation (1)</td></tr><tr><td>$N_{p,y}$</td><td>Number of premises type p with at least one project technology in year y</td><td>Sales database</td></tr><tr><td>$HN_{p,y}$</td><td>Number of individuals per premises type p in year y</td><td>Project survey, for ex ante estimation, result of 3.9 from baseline survey shall be used.</td></tr><tr><td>$DP_{p,y}$</td><td>Days the project technology is present for end-users in the premises p in year y</td><td>To be monitored, for ex ante estimation, $D_{p,y}=365$</td></tr><tr><td>$U_{p,y}$</td><td>Usage rate of the project technology by premises type p during year y.</td><td>To be monitored, for ex ante estimation, $U_{p,y}=80\%^{32}$</td></tr><tr><td>$M_{q,y}$</td><td>Ongoing water quality indicated as the fraction of the samples that pass microbial quality standard requirements specified in relevant microbial quality</td><td>Water quality test, for ex ante estimation, $M_{q,y}=100\%$ based on the manufacturer's specifications of the project technologies</td></tr></table>	Parameters	Description	Source/Value	Project outcome of SDG 6	Total number of person.days served with satisfactory level of safe water	Calculated based on equation (1)	$N_{p,y}$	Number of premises type p with at least one project technology in year y	Sales database	$HN_{p,y}$	Number of individuals per premises type p in year y	Project survey, for ex ante estimation, result of 3.9 from baseline survey shall be used.	$DP_{p,y}$	Days the project technology is present for end-users in the premises p in year y	To be monitored, for ex ante estimation, $D_{p,y}=365$	$U_{p,y}$	Usage rate of the project technology by premises type p during year y .	To be monitored, for ex ante estimation, $U_{p,y}=80\%^{32}$	$M_{q,y}$	Ongoing water quality indicated as the fraction of the samples that pass microbial quality standard requirements specified in relevant microbial quality	Water quality test, for ex ante estimation, $M_{q,y}=100\%$ based on the manufacturer's specifications of the project technologies
Parameters	Description	Source/Value																						
Project outcome of SDG 6	Total number of person.days served with satisfactory level of safe water	Calculated based on equation (1)																						
$N_{p,y}$	Number of premises type p with at least one project technology in year y	Sales database																						
$HN_{p,y}$	Number of individuals per premises type p in year y	Project survey, for ex ante estimation, result of 3.9 from baseline survey shall be used.																						
$DP_{p,y}$	Days the project technology is present for end-users in the premises p in year y	To be monitored, for ex ante estimation, $D_{p,y}=365$																						
$U_{p,y}$	Usage rate of the project technology by premises type p during year y .	To be monitored, for ex ante estimation, $U_{p,y}=80\%^{32}$																						
$M_{q,y}$	Ongoing water quality indicated as the fraction of the samples that pass microbial quality standard requirements specified in relevant microbial quality	Water quality test, for ex ante estimation, $M_{q,y}=100\%$ based on the manufacturer's specifications of the project technologies																						

³² Refer to the registered GS project Aqua Clara Water Filtration Program in Kenya (GS 1078). <https://registry.goldstandard.org/projects/details/59>

			<table><tr><td></td><td>standard for drinking water of Kenya.</td><td></td></tr></table> The net benefit of SDG6 = Project outcome of SDG6 – Baseline outcome of SDG6		standard for drinking water of Kenya.	
	standard for drinking water of Kenya.					
SDG7 Affordable and Clean Energy	The number of water purifiers sold each year	In baseline scenario, no project water purifiers will be used. Therefore, the baseline outcome is 0.	The number of water purifiers sold will be based on the sales database and cross checked by the purchase receipts. The net benefit of SDG7 = Project outcome of SDG7- Baseline outcome of SDG7			
SDG 8 Decent Work and Economic Growth	1) Number of males and females employed by the project.	In baseline situation, no new job created. Therefore, the baseline outcome benefit is 0.	In project situation, the number of jobs created for males and females will be based on the labor contact and employee payroll. Net benefit of SDG8=Project outcome of SDG8 – Baseline outcome of SDG8			
	2) Percentage of employees with salaries paid at par with the average wage of Kenya.	In baseline situation, no new job created. Therefore, the baseline outcome benefit is 0.	The project implementer will check the related wage laws of Kenya and record the number of employees that are paid at par with the average wage of Kenya³³. The salary of this project employee will be based on the employee payroll. The net benefit of this parameter can be calculated as: The number of employees that are paid at par with the average wage of Kenya/The number of jobs created*100%. Net benefit of SDG8=Project outcome of SDG8 – Baseline outcome of SDG8			
SDG 13 Climate Action	GHG emission reductions	<i>Methodology for Emission Reductions from Safe Drinking Water Supply</i> (Version 1.0) provides specific surveys, calculations for baseline and project emissions as well as guidance on applicable surveys and tests.				

Specific Calculation Method for Outcome of SDG 13 Baseline emissions

As per section 3.6 of the applied GS methodology, baseline emissions can be calculated as follows:

$$BE_y = EF_b * (1 - C_b - X_{cleanboil,y}) * Q_y * M_{q,y} \quad (2)$$

Where:

³³ For ex estimated, the pre-calculation refers to the average salary in Kenya in 2019, Kenya's Average Wage Earnings data was reported at 778,248 KES in 2019. (<https://www.ceicdata.com/en/kenya/average-wage-earnings-by-sector-and-industry-international-standard-of-industrial-classification-rev-4/average-wage-earnings>). During the monitoring period, it should refer to the latest public data that is available.

Parameters	Description	Source/value
BE_y	Baseline emissions from the use of fuel to obtain safe water in the baseline (tCO ₂ e)	
EF_b	Emission factor for the use of fuel to obtain safe water in the baseline (tCO ₂ e/L)	Calculated as following Equation (3) and (4)
C_b	Proportion of project households who in the baseline were already using a safe water supply that did not require boiling it (%)	Baseline survey, 20.0%
$X_{cleanboil,y}$	Proportion of project households that boil safe water in the project year y (%)	Usage survey, for ex-ante estimation, it assumes a value of 0%. The actual value will be monitored.
Q_y	Quantity of safe drinking water provided by the project in year y (L)	Calculated based on Method 2 for household water treatment technologies.
$M_{q,y}$	Modifier for the water quality in year y	For ex-ante estimation, it assumes a value of 100%. The actual value will be determined based on water quality test.

$$EF_b = SE_{w,b,y} * \sum_f (x_f * (EF_{b,f,CO_2} * f_{NRB,f,y} + EF_{b,f,nonCO_2})) \div 10^9 \quad (3)$$

As per results of baseline survey, among those who treat drinking water by boiling, the main fuel used are non-renewable biomass of firewood, charcoal and LPG. EF_b can be calculated as the following equation.

$$EF_b = SE_{w,b,y} * (x_{firewood} * (EF_{b,firewood,CO_2} * f_{NRB} + EF_{b,firewood,nonCO_2}) + x_{charcoal} * (EF_{b,charcoal,CO_2} * f_{NRB} + EF_{b,firewood,nonCO_2}) + x_{LPG} * (EF_{b,LPG,CO_2} + EF_{b,LPG,nonCO_2})) \div 10^9 \quad (4)$$

Where:

Parameters	Description	Source/value
EF_b	Emission factor for the use of fuel to obtain safe water in the baseline (tCO ₂ e/L)	Calculated as Equation (3) and (4), Makueni (2.713*10 ⁻⁴), Trans Nzoia (2.374*10 ⁻⁴), Kirinyaga (2.848*10 ⁻⁴). Bungoma (2.356*10 ⁻⁴) and Nakuru (1.493*10 ⁻⁴)
$SE_{w,b,y}$	Specific energy required to boil water (kJ/L)	Calculated as Equation (5), Makueni (2,793.52), Trans Nzoia (2,440.91), Kirinyaga (3,207.38). Bungoma (2,465.08) and Nakuru (1,576.60)
$x_{firewood}$	Proportion of firewood used in the baseline (fraction)	Baseline survey, Makueni (81.3%), Trans Nzoia (78.3%), Kirinyaga (93.1%). Bungoma (79.7%) and Nakuru (56.7%)

$x_{charcoal}$	Proportion of charcoal used in the baseline (fraction)	Baseline survey, Makueni (18.8%), Trans Nzoia (19.6%), Kirinyaga (6.9%). Bungoma (17.4%) and Nakuru (21.6%)
x_{LPG}	Proportion of LPG used in the baseline (fraction)	Baseline survey, Makueni (0%), Trans Nzoia (2.2%), Kirinyaga (0%). Bungoma (2.9%) and Nakuru (21.6%)
$EF_{b,firewood,CO_2}$	CO ₂ emission factor from use of firewood (tCO ₂ /TJ)	IPCC 2006 default value for wood (Table 1.4 of Chapter 1 Introduction), 112 tCO ₂ /TJ
$EF_{b,firewood,nonCO_2}$	Non-CO ₂ emission factor arising from use of firewood (tCO ₂ e/TJ)	<i>Methodology for Emission Reductions from Safe Drinking Water Supply</i> (Version 1.0), 9.46t/TJ
$EF_{b,charcoal,CO_2}$	CO ₂ emission factor from use of charcoal (tCO ₂ /TJ)	<i>Methodology for Emission Reductions from Safe Drinking Water Supply</i> (Version 1.0), 165.22 tCO ₂ /TJ (includes charcoal production emissions)
$EF_{b,charcoal,nonCO_2}$	Non-CO ₂ emission factor arising from use of charcoal (tCO ₂ e/TJ)	<i>Methodology for Emission Reductions from Safe Drinking Water Supply</i> (Version 1.0), 44.83t/TJ (includes production emissions of CH ₄ and N ₂ O)
EF_{b,LPG,CO_2}	CO ₂ emission factor from use of LPG (tCO ₂ /TJ)	IPCC 2006 default value for wood (Table 1.4 of Chapter 1 Introduction), 63.1 tCO ₂ /TJ
$EF_{b,LPG,nonCO_2}$	Non-CO ₂ emission factor arising from use of LPG (tCO ₂ e/TJ)	Calculated as following Equation (6), 0.1665t/TJ
$f_{NRB,f,y}$	Fractional non-renewability status of woody biomass fuel during year y , in case the baseline fuel is biomass or charcoal	Baseline survey, 66.4% and fixed for the first crediting period.
f	Index for baseline fuel types	Baseline survey, the baseline fuel types are firewood, charcoal and LPG.

The specific energy required to boil water using the baseline technology ($SE_{w,b,y}$) is determined as follows, by calculating the energy input required to obtain 1 L of boiling water, including boiling and vaporization losses.

$$SE_{w,b,y} = 360.83/\eta_{wb} \quad (5)$$

Where:

360.83	Default amount of energy required to boil 1 L of water from a first principles approach
η_{wb}	Efficiency of the baseline water boiling (%). Weighted average of baseline stove types. Default efficiency: 10% for three-stone stove, 20% for conventional stove, 30% for improved stove. 50% for LPG stove based on manufacturer specification. Based on the results of baseline survey, η_{wb} of each County: Makueni (12.9%), Trans Nzoia (14.8%), Kirinyaga (11.3%). Bungoma (14.6%) and Nakuru (22.9%)

According to IPCC 1996³⁴, the calculation of non-CO₂ emission factors needs to fill each other of gases (CH₄, N₂O, NO_x, CO and NMVOC). As per table 2.5 of Chapter 2 Stationary Combustion of IPCC 2006, LPG Emission CH₄ Conversion Factor is 0.005t/TJ, LPG Emission N₂O Conversion Factor is 0.0001t/TJ, the non-CO₂ emission factors of LPG is the emission conversion factors multiply the corresponding GHG potential, i.e.,

$$EF_{b,LPG,nonCO2} = \text{LPG Emission CH}_4 \text{ Conversion Factor} * GWP_{CH4} + \text{LPG Emission N}_2\text{O Conversion Factor} * GWP_{N2O} \quad (6)$$

As per Table 8.7 of 2013 *Climate Change 2013 The Physical Science Basis. Contribution of Working Group I to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change* (IPCC)³⁵, GWP₁₀₀ of CH₄ is 28 (no climate-carbon feedbacks, cc fb) and GWP₁₀₀ of N₂O is 265 (no climate-carbon feedbacks, cc fb).

Based on Equation (5), $EF_{b,LPG,nonCO2} = 0.005\text{t/TJ} * 28 + 0.0001\text{t/TJ} * 265 = 0.1665\text{t/TJ}$

Next, calculate the baseline emissions.

The quantity of safe drinking water provided by the project (Q_y) can be calculated using Method 2, which applies to projects using household water treatment technologies.

$$Q_y = \sum_p N_{p,y} * U_{p,y} * QPW_{hh,p,y} * DP_{p,y} \quad (7)$$

Where:

Parameters	Description	Source/value
$N_{p,y}$	Number of premises type p with at least one project technology in year y	Sales database
$U_{p,y}$	Usage rate of the project technology by premises type p during year y (%)	Usage rate survey. For ex-ante estimation, $U_{p,y} = 80\%$ ³⁶
$QPW_{hh,p,y}$	Volume of drinking water per premises p per day in year y (L)	Calculated as Equation (8), 15

³⁴ <https://www.ipcc-nggip.iges.or.jp/public/gl/guidelin/ch1wb1.pdf>

³⁵ <https://www.ipcc.ch/report/ar5/wg1/>

³⁶ Refer to the registered GS project Aqua Clara Water Filtration Program in Kenya (GS 1078). <https://registry.goldstandard.org/projects/details/59>

$DP_{p,y}$	Days the project technology is present for end-users in the premises p in year y	The actual value will be determined based on the sales database. For ex-ante estimation, it assumes a value of 365.
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The volume of drinking water per household per day is determined by considering whether the capacity of the project device is sufficient to provide at least the default amount of drinking water, as follows.

$$QPW_{hh,p,y} = \min \left((q_i * t_{p,y} * DN_{p,y}), (QPW_p * HN_{p,y}) \right) \quad (8)$$

Where:

Parameters	Description	Source/value
q_i	Capacity of the household water treatment technology (L/h)	Manufacturer specifications, 3
$t_{p,y}$	Usage time of the project technology by premises type p in year y (h/day)	Option 3 Default value of 5 hours will be used.
$DN_{p,y}$	Average number of individual project technologies in each project premises type p in year y	The actual value will be determined based on sales database. For ex-ante estimation, it assumes a value of 1.
$HN_{p,y}$	Number of individuals per premises type p in year y	The actual value will be determined based on project survey. For ex-ante estimation, result of 3.9 from baseline survey shall be used.
QPW_p	Volume of drinking water per person per day for premises type p (L)	The actual value will be determined based on Water Consumption Field Tests. For ex ante estimation, the result of 4.3 L/person/day from baseline survey will be used.

Project emissions

As per section 3.7 of the applied GS methodology, project emissions may result from the operation of new low-emission water treatment technologies. The Multi-Layer water purifiers introduced by the project require no energy input or consumables, which will offer an affordable, long-term and zero emission solution for households that generally consume unsafe drinking water. No fossil fuel or electricity will be used in the project activity, project emissions (PE_y) is 0 tCO_{2e}.

Leakage emissions (LE_y)

As per section 3.8 of the applied GS methodology, leakage relating to the non-renewable woody biomass shall be assessed and other types of leakage are excluded for simplification. Leakage emissions are investigated as follows.

First, the project developer must evaluate, ex-ante, the following potential sources of leakage and provide an evidence-based description and preliminary quantification of each potential source and its relevance for the project:

- a. Members of the population who do not participate in the project, and previously used lower emitting energy sources, instead use the nonrenewable biomass saved under the project activity.

The volume of water treated by boiling in the baseline consumes a fractional portion of the biomass used by families. Fuel wood previously used was sourced or obtained individually and fuel wood consumption is a function of market forces of demand and supply. Biomass is currently non-renewable and expensive for families. It is highly unlikely that any biomass saved by the project activity will significantly reduce biomass costs outside the project boundary. Therefore, the project proponent assigns a value of 0tCO_{2e} to this leakage parameter.

- b. The project significantly reduces the NRB fraction within an area where other GHG mitigation project activities account for NRB fraction in their baseline scenario.

As per section 3.6.2 of the applied GS methodology, the amount of energy required to boil 1 L of water is 360.83 kJ/L. The annual average quantity of safe water provided by the project is 564,888,600 L. Based on the results of baseline survey, the weighted average efficiency of the baseline stove for water boiling is 16.1%. Annual average thermal energy savings is $360.83 \times 564,888,600 / 16.1\% / 1,000,000 = 1,266,017.103 \text{ GJ} = 1.266 \text{ PJ}$, which amounts to a 0.15%³⁷ change in the NRB fraction of Kenya. It is an insignificant impact on the NRB fraction of Kenya.

- c. The project population compensates for loss of the space heating effect of water boiling by adopting some other form of space heating or by retaining some baseline wood fuel-burning practices.

In Kenya, households rarely use stoves for heating as the average daytime temperature is between 20°C to 28°C all year long³⁸. Therefore, it is unlikely that the project activity will result in increased use of biomass for space heating effects. Thus, the project proponent assigns a value of 0 tCO_{2e} to this leakage parameter.

³⁷ As per the results of NRB assessment, biomass fuels consumption in Kenya in 2017 was 724 PJ, in which sustainable potential of solid biomass was 243 PJ. Therefore, DRB is 243 PJ, NRB equals to $724 - 243 = 481 \text{ PJ}$. $f_{NRB} = 481 / 724 = 66.4\%$. Annual average thermal energy savings due to the project activity is 1.266PJ. Annual NRB and biomass fuels consumption may raise to $481 + 1.266 = 482.266 \text{ PJ}$ and $724 + 1.266 = 725.266 \text{ PJ}$ respectively. f_{NRB} may change to $482.266 / 725.266 = 66.5\%$. The impact of the project activity on the NRB fraction nationally can be calculated as $66.5\% / 66.4\% - 1 = 0.15\%$.

³⁸ <https://www.safaribookings.com/kenya/climate>

In summary, Leakage risks of the project activity deemed very low and can be ignored. Leakage emissions (LE_y)=0 tCO₂e. Leakage investigation will be conducted every two years. The parameters of leakage emissions are included in the monitoring plan. Refer to section B.7 of the PDD for more details.

GHG Emission Reductions

$$ER_y = BE_y - PE_y - LE_y \quad (9)$$

Parameters	Description	Source/value
ER_y	Emission reductions in year y (tCO ₂ e/yr)	Calculated as Equation (9)
BE_y	Baseline emissions in year y (tCO ₂ e/yr)	Calculated as Equation (2) (3), (5), (7) and (8)
PE_y	Project emissions in year y (tCO ₂ e/yr)	0 tCO ₂ e
LE_y	Leakage emissions in year y (t CO ₂ e/yr)	0 tCO ₂ e

B.6.2 Data and parameters fixed ex ante

Related to water quality

Parameter ID	SDWS 2
Data/parameter	Project technology description
Unit	N/A
Description	The Multi-Layer water purifiers are produced by Zhejiang Duanqi Renewable Energy Co., Ltd., which consist of 8 step filtration system and can purify water 3 liter/hour. The lifespan of the water purifiers are over 15 years and the dome ceramic, carbon cartridge and mineral stone should be changed when purify every 10,000 L water. The project equipment provides safe drinking water that are free from pathogens and elevated levels of toxic substances. Refer to section A.3 for more details. In addition, the water purifier involved in this project obtain the pre-export verification of conformity by KEBS (Kenya bureau of standards). Moreover, according to the water quality test report by the KEBS, the purified water quality of the product meets the standards of KS EAS 153: 2018 East African Standard Specification for Packaged Drinking Water.
Source of data	Manufacturer specifications and the pre-export verification of conformity.
Value(s) applied	N/A
Choice of data or Measurement methods and procedures	The manufacturer specifications, the pre-export verification of conformity and the water quality test report provide accurate evidence of compliance with relevant standards. It also can be known from the sales contract that the water purifier is also presented with a filter element as a spare when it is sold. When

	the water purifier has been used for less than 22 months, when the user needs to replace the filter element, a request can be sent to the project owner, and the user can use a spare filter element during the time period when the project owner comes to replace it. Of course, after receiving the request, the project owner will immediately go to the user's home to replace the core filter. When the water purifier has been used for 22 months, the project owner will replace the alternative materials of all the water purifier sold.
Purpose of data	The Project technology of the water purifier involved in this project meet the relevant standard and the product is eligible products.
Additional comment	N/A

Parameter ID	<i>SDWS 4</i>
Data/parameter	<i>Regulatory framework for safe water supply</i>
Unit	N/A
Description	Water scarcity in various regions of Kenya has been an issue for decades. The government has issued some water resources related policies and regulations. The <i>Water Act 2016</i> provides for the regulation, management and development of water resources and water and sewerage services in line with the Constitution. The <i>National Water Quality Management Strategy (NWQMS)</i> issued by the Ministry of Water and Irrigation to improve water quality. The <i>Kenya National Water Master Plan 2030</i> developed by Ministry of Environment, Water and Natural Resources recognizes water as an enabler in socio-economic development and aims to improve water and sanitation availability and access to all by 2030. The implementation of the project is fully in line with the relevant water resources policies of Kenya.
Source of data	National authorities
Value(s) applied	N/A
Choice of data or Measurement methods and procedures	National authorities provide the powerful regulatory framework for safe water supply
Purpose of data	The Kenyan government attaches great importance to the issue of water resources. This project could bring water benefits to the Kenyan people. So, the implementation of the project is fully in line with the relevant water resources policies of Kenya.
Additional comment	N/A

Parameter ID	<i>SDWS 5</i>
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Data/parameter	<i>Water sources in the project boundary</i>
Unit	N/A
Description	The water source in the project boundary and whether they are used for drinking water, and for all that are used for drinking water, classify them as improved and unimproved water source.
Source of data	Baseline study conducted from 10/10/2020 to 20/12/2020.
Value(s) applied	0.59% (3HH/505HH) of the households are available of piped water. 6.7% (34HH/505HH) of the households take water from boreholes or protected wells. As per Annex 2 of the applied GS methodology, piped water and water from boreholes or protected wells belong to improved sources of drinking water. For the rest of 92.7% (468HH/505HH) households, the most common sources of water are from unprotected well, unprotected spring and surface water, which belong to unimproved sources of drinking water.
Choice of data or Measurement methods and procedures	The baseline study conducted from 10/10/2020 to 20/12/2020 provided the water source in project boundary.
Purpose of data	From the baseline survey, the most people in Kenya can't obtained the safety water from improved water source, so the implementation of this project is important for the safety of drinking water for Kenyans. Demonstrated the reasonableness of project implementation.
Additional comment	N/A

SDG13

Parameter ID	<i>SDWS 6</i>
Data/parameter	<i>Stove technologies used in the project boundary</i>
Unit	N/A
Description	The proportion of different stove types used in premises in the geographical area of the project.
Source of data	Baseline survey conducted from 10/10/2020 to 20/12/2020.
Value(s) applied	The stove used to boil water in baseline scenario is three-stone stoves (67.2%, 223HH/332HH), Conventional stoves (19.3%, 64HH/332HH), improved stoves (6.3%, 21HH/332HH) and LPG stoves (7.2%, 24HH/332HH). The project is implemented in Makueni, Trans Nzoia, Kirinyaga, Bungoma and Nakuru Counties of Kenya, the proportion of different stove types are defined for each county. Refer to section B.4.1 of the PDD for more details
Choice of data or Measurement methods and procedures	The baseline study conducted from 10/10/2020 to 20/12/2020 provided the proportion of different stove types used in premises in the geographical area of the project.
Purpose of data	To calculate the baseline emissions.

Additional comment	The classification shall consider at least the following categories of stoves types: - Three-stone fire or a conventional system for woody biomass lacking improved combustion air supply mechanism and flue gas ventilation system; - other conventional systems using woody biomass; - improved cookstoves ($\geq 20\%$ thermal efficiency); and - fossil fuel combusting systems. If the project is implemented in different geographical locations, then the proportion of different stove types shall be defined for each location, where relevant.
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Parameter ID	<i>SDWS 7</i>
Data/parameter	<i>Expected technical life of project technology</i>
Unit	Operational lifespan of the water purifiers are more than 15 years
Description	As per manufacturer specifications of the water purifiers, the dome ceramic should be cleaned every 7 days with toothbrush, the outside of carbon cartridge should be cleaned every 10 days and the dome ceramic, carbon cartridge and mineral stone should be changed when purify every 10,000 L water (about 22 months).
Source of data	Manufacturer specifications
Value(s) applied	More than 15 years
Choice of data or Measurement methods and procedures	The manufacturer specifications provided the accuracy information of expected technical life of an individual project technology and the details include both technology/device life and filter life. In order to enable the water purifier to last for 15 years of technical life, the project owner will regularly replace replaceable materials and repair other parts. If the core filters have reached the replacement time and the user has not requested replacement, the project owner will ask the user whether it is time to replace the core filters through on-site investigation or telephone communication and ask the user to replace the core filters free of charge. During the use time, the user takes the initiative to report a request to replace the core filters, then the project owner will also come to replace it for free and fill in the replacement record information in the "replacement and maintenance record" form.
Purpose of data	The expected technical life of project technology meets the total length of crediting period.
Additional comment	N/A

Parameter ID	SDWS 8																															
Data/parameter	$X_{firewood}/X_{charcoal}/X_{LPG}$																															
Unit	Percentage of firewood, charcoal and LPG																															
Description	The proportion of each different household cooking fuel (firewood, charcoal and LPG) used in the project boundary																															
Source of data	Baseline survey conducted from 10/10/2020 to 20/12/2020.																															
Value(s) applied	<table><tr><td>County</td><td>$X_{firewood}$</td><td>$X_{charcoal}$</td><td>X_{LPG}</td></tr><tr><td>Makueni</td><td>81.3</td><td>18.8</td><td>0</td></tr><tr><td>Trans Nzoia</td><td>78.3</td><td>19.6</td><td>2.2</td></tr><tr><td>Kirinyaga</td><td>93.1</td><td>6.9</td><td>0</td></tr><tr><td>Bungoma</td><td>79.7</td><td>17.4</td><td>2.9</td></tr><tr><td>Nakuru</td><td>56.7</td><td>21.6</td><td>21.6</td></tr><tr><td>Total</td><td>75.9</td><td>16.9</td><td>7.2</td></tr></table>				County	$X_{firewood}$	$X_{charcoal}$	X_{LPG}	Makueni	81.3	18.8	0	Trans Nzoia	78.3	19.6	2.2	Kirinyaga	93.1	6.9	0	Bungoma	79.7	17.4	2.9	Nakuru	56.7	21.6	21.6	Total	75.9	16.9	7.2
County	$X_{firewood}$	$X_{charcoal}$	X_{LPG}																													
Makueni	81.3	18.8	0																													
Trans Nzoia	78.3	19.6	2.2																													
Kirinyaga	93.1	6.9	0																													
Bungoma	79.7	17.4	2.9																													
Nakuru	56.7	21.6	21.6																													
Total	75.9	16.9	7.2																													
Choice of data or Measurement methods and procedures	505 households participated the baseline survey (101 households for each County). 332 households (Makueni 48, Trans Nzoia 46, Kirinyaga 72, Bungoma 69 and Nakuru 97) treat unsafe drinking water by boiling. The questionnaire shown in Table B4.1 are adapted to capture the information of fuel types and stove types used to boil water. The data was collected through paper surveys, which were then entered into an excel file to be used to analyze the data. Refer to section B4.1 of this PDD for more details.																															
Purpose of data	To demonstrate contribution to SDG 13.																															
Additional comment	The proportion of different fuel types are defined for each County. According to applied methodology, the percentages applied shall be cross-checked against at least one other source on the list. For cross-check purposes, sources applied may be up to 5 years old. In this project the cross-check document is <i>water survey report</i> approved by National Environment Management Authority, which belongs to credible published literature for project region. According to <i>water survey report</i> approved by National Environment Management Authority in Turkana, Makueni, Kirinyaga, Mandera, Tran Nzoia, Bungoma and Nakuru County, the main fuel used for boiling water were firewood and products of wood 77.1%, charcoal 17.1% and LPG 5.8%. So, consider conservative principles, the percentage of firewood, charcoal and LPG is 75.9%, 16.9% and 7.2% in the calculation of baseline emissions.																															

Parameter ID	SDWS 9
Data/parameter	$EF_{b,firewood,CO2}$
Unit	tCO ₂ /TJ
Description	CO ₂ emissions factor of the fuel (firewood) that it substituted or reduced
Source of data	IPCC 2006 default value (Table 1.4 of Chapter 1 Introduction)

Value(s) applied	112
Choice of data or Measurement methods and procedures	IPCC default values provide an accurate and conservative estimate of emissions reduction from various fuel sources.
Purpose of data	To demonstrate contribution to SDG 13.
Additional comment	N/A

Parameter ID	<i>SDWS 9</i>
Data/parameter	$EF_{b,charcoal,CO2}$
Unit	tCO ₂ /TJ
Description	CO ₂ emissions factor of the fuel (charcoal) that it substituted or reduced
Source of data	<i>Methodology for Emission reductions from safe drinking water supply (Version 1.0)</i>
Value(s) applied	165.22 (includes charcoal production emissions)
Choice of data or Measurement methods and procedures	IPCC default values provide an accurate and conservative estimate of emissions reduction from various fuel sources.
Purpose of data	To demonstrate contribution to SDG 13.
Additional comment	N/A

Parameter ID	<i>SDWS 9</i>
Data/parameter	$EF_{b,LPG,CO2}$
Unit	tCO ₂ /TJ
Description	CO ₂ emissions factor of the fuel (LPG) that it substituted or reduced
Source of data	IPCC 2006 default value (Table 1.4 of Chapter 1 Introduction)
Value(s) applied	63.1
Choice of data or Measurement methods and procedures	IPCC default values provide an accurate and conservative estimate of emissions reduction from various fuel sources.
Purpose of data	To demonstrate contribution to SDG 13.
Additional comment	N/A

Parameter ID	<i>SDWS 10</i>
Data/parameter	$EF_{b,firewood,nonCO2}$
Unit	t/TJ
Description	Non-CO ₂ emissions factor of the firewood that is substituted or reduced

Source of data	Table 2.5 of Chapter 2 Stationary Combustion (IPCC 2006) and Table 8.7 of 2013 Climate Change 2013 The Physical Science Basis. Contribution of Working Group I to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change (IPCC).
Value(s) applied	9.46 (0.3t/TJ*28+ 0.004t/TJ*265)
Choice of data or Measurement methods and procedures	Determined as per IPCC default figures.
Purpose of data	To demonstrate contribution to SDG 13.
Additional comment	N/A

Parameter ID	<i>SDWS 10</i>
Data/parameter	<i>EF_{b,charcoal,nonCO2}</i>
Unit	t/TJ
Description	Non-CO ₂ emissions factor of the firewood that is substituted or reduced
Source of data	<i>Methodology for Emission reductions from safe drinking water supply (Version 1.0)</i>
Value(s) applied	44.83 (includes production emissions of CH ₄ and N ₂ O)
Choice of data or Measurement methods and procedures	Determined as per IPCC default figures.
Purpose of data	To demonstrate contribution to SDG 13.
Additional comment	N/A

Parameter ID	<i>SDWS 10</i>
Data/parameter	<i>EF_{b,LPG,nonCO2}</i>
Unit	t/TJ
Description	Non-CO ₂ emissions factor of the LPG that is substituted or reduced
Source of data	Table 2.5 of Chapter 2 Stationary Combustion (IPCC 2006) and Table 8.7 of 2013 Climate Change 2013 The Physical Science Basis. Contribution of Working Group I to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change (IPCC).
Value(s) applied	0.1665 (0.005t/TJ*28+ 0.0001t/TJ*265)
Choice of data or Measurement methods and procedures	Determined as per IPCC default figures.
Purpose of data	To demonstrate contribution to SDG 13.
Additional comment	N/A

Parameter ID	SDWS 11													
Data/parameter	η_{wb}													
Unit	%													
Description	Weighted average efficiency of the baseline water boiling. Calculate the weighted average of the water boiling efficiency in the project boundary using the proportion of different stove types used and the stove efficiencies.													
Source of data	Baseline Survey													
Value(s) applied	<table><tr><td>County</td><td>η_{wb}</td></tr><tr><td>Makueni</td><td>12.9</td></tr><tr><td>Trans Nzoia</td><td>14.8</td></tr><tr><td>Kirinyaga</td><td>11.3</td></tr><tr><td>Bungoma</td><td>14.6</td></tr><tr><td>Nakuru</td><td>22.9</td></tr></table>		County	η_{wb}	Makueni	12.9	Trans Nzoia	14.8	Kirinyaga	11.3	Bungoma	14.6	Nakuru	22.9
County	η_{wb}													
Makueni	12.9													
Trans Nzoia	14.8													
Kirinyaga	11.3													
Bungoma	14.6													
Nakuru	22.9													
Choice of data or Measurement methods and procedures	505 households participated the baseline survey (101 households for each County). 332 households (Makueni 48, Trans Nzoia 46, Kirinyaga 32, Bungoma 69 and Nakuru 137) treat unsafe drinking water by boiling. The questionnaire shown in Table B4.1 are adapted to capture the information of fuel types and stove types used to boil water. The data was collected through paper surveys, which were then entered into an excel file to be used to analyze the data. Refer to section B4.1 of this PDD for more details. Default efficiency values for different type of stoves: three-stone stoves 10%, conventional stoves 20%, improved stoves 30%. As per manufacturer specifications of the baseline LPG stoves, the highest efficiency is 50%.													
Purpose of data	To demonstrate contribution to SDG 13.													
Additional comment	N/A													

Parameter ID	SDWS 12	
Data/parameter	C_b	
Unit	%	
Description	Proportion of project households who in the baseline were already using a safe water supply that did not require boiling.	
Source of data	Baseline Survey conducted from 10/10/2020 to 20/12/2020.	
Value(s) applied	20.0	
Choice of data or Measurement methods and procedures	The percentage of households who are available of safe drinking water before project activity is 7.3% (37HH/505HH). 78.4% (396HH/505HH) treat unsafe water before drinking and 14.3% (72HH/505HH) drink unsafe water directly without any treatment. Data and information collected from the questionnaires show that all the households surveyed are	

	involved in water treatment for domestic consumption. Among those who treat water, 83.8% (332HH/396HH) of them boil water before drinking and 16.2% (64HH/396HH) use gravity household water filters, ultraviolet radiation treatment, reverse osmosis, chlorine tablets and other technologies. Therefore, C_b can be calculated as $7.3\% + 78.4\% \times 16.2\% = 20.0\%$.
Purpose of data	To demonstrate contribution to SDG 13.
Additional comment	According to applied methodology, the percentages applied shall be cross-checked against at least one other source on the list. For cross-check purposes, sources applied may be up to 5 years old. In this project the cross-check document is <i>Water Survey Report</i> approved by National Environment Management Authority, which belongs to Credible published literature for project region. According to <i>Water Survey Report</i> approved by National Environment Management Authority, in Turkana, Makueni, Kirinyaga, Mandera, Tran Nzoia, Bungoma and Nakuru County, 0.43% (2HH/462HH) of the households were available of piped water or bottled water. 5.8% (27HH/462HH) of the households got water from boreholes or protected wells. Piped water, bottled water and water from boreholes or protected wells were improved sources for drinking (6.3%, 29HH/462HH). Among those who were not available of safe water (93.7%, 433HH/462HH), 12.9% (56HH/433HH) will treat unsafe water before drinking by chlorine tablets, water filters or similar technologies, 75.5% (327HH/433HH) will drink boiled water and the rest of them (18.2%, 79HH/433HH) will drink unsafe water directly. Therefore, the proportion of households who were already using a safe water supply that did not require boiling it can be calculated as $6.3\% + 12.9\% \times 93.7\% = 18.4\%$, which is less than 20.0%.

Parameter ID	SDWS 13
Data/parameter	q_i
Unit	Liters per hour
Description	Capacity of the household water treatment technology
Source of data	Manufacturer specifications
Value(s) applied	3
Choice of data or Measurement methods and procedures	Manufacturer specifications provides the capacity, lifetime and water quality of the household water treatment technology.
Purpose of data	To demonstrate contribution to SDG 13.
Additional comment	N/A

Parameter ID	SDWS 21
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Data/parameter	$f_{NRB,f,y}$
Unit	%
Description	Fractional non-renewability status of woody biomass fuel during year y , in case the baseline fuel is biomass or charcoal
Source of data	Determined by CDM Tool 30 Calculation of the fraction of non-renewable biomass (Version 03.0)
Value(s) applied	66.4
Choice of data or Measurement methods and procedures	Since it is difficult to estimate the locations and extent of areas from which woody biomass fuel used by the project is collected, the project proponent aggregates all reachable woody biomass fuel collection areas within Kenya and apply a single fraction derived from all collection areas in the country. According to <i>East African Civil Society for Sustainable Energy and Climate Action</i> (EASE & CA) published on August 2020³⁹, biomass fuels consumption in Kenya in 2017 was 724 PJ⁴⁰, in which sustainable potential of solid biomass was 243 PJ. Therefore, RB is 243 PJ, NRB equals to $724-243=481$ PJ. $f_{NRB}=481/ (481+243) =66.4\%$ The f_{NRB} of 66.4% is fixed based on the results of the NRB assessment for the first crediting period.
Purpose of data	To demonstrate contribution to SDG 13.
Additional comment	N/A

B.6.1. Ex ante estimation of SDG Impact

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SDG 6 Clean Water and Sanitation

For SDG6, the monitoring indicator is Total number of person.days served with satisfactory level of safe water and the calculation formula is as follows:

The Project outcome of SDG 6 = $N_{p,y} * HN_{p,y} * DP_{p,y} * U_{p,y} * M_{q,y}$

where,

- $N_{p,y}$ Number of premises type p with at least one project technology in year y
- $HN_{p,y}$ Number of individuals per premises type p in year y , Project survey, for ex ante estimation, result of 3.9 from baseline survey shall be used.
- $DP_{p,y}$ Days the project technology is present for end-users in the premises p in year y , To be monitored, for ex ante estimation, $DP_{p,y}=365$

³⁹ https://www.inforse.org/africa/pdfs/Pub_100-Renewable-Energy-Plan-for-Kenya-by-2050-12-08-2020.pdf

⁴⁰ 1PJ=10¹⁵J.

$U_{p,y}$	Usage rate of the project technology by premises type p during year y . To be monitored, for ex ante estimation, $U_{p,y}=80\%$ ⁴¹
$M_{q,y}$	Ongoing water quality indicated as the fraction of the samples that pass microbial quality standard requirements specified in relevant microbial quality standard for drinking water of Kenya. Water quality test, for ex ante estimation, $M_{q,y}=100\%$ based on the manufacturer's specifications of the project technologies

The total number of person.days served with satisfactory level of safe water of outcomes for SDG 6 are shown as the following table.

Year	Project outcome of SDG 6
1	553,423,842 person.days
2	694,662,218 person.days
3	732,133,624 person.days
4	740,780,871 person.days
5	743,490,342 person.days
Total	3,464,490,896 person.days

SDG 7 Affordable and Clean Energy

The number of water purifiers sold will be based on the sales database and cross checked by the purchase receipts. According to the sales plan during the first crediting period, the outcomes for SDG 7 are shown as the following table. In order to cover more households with high-quality water purifiers at a lower price, it is restricted in the project that each household can only buy one set of water purifier.

Table B6.1 Water Purifiers Sales Plan during the First Crediting Period

Year	Estimated number of water purifiers sold in year y (sets)					
	Makueni	Trans Nzoia	Kirinyaga	Bungoma	Nakuru	Total
1	80,000	75,000	65,000	95,000	165,000	480,000
2	20,000	18,000	15,500	24,000	45,000	122,500
3	5,000	4,500	4,000	6,500	12,500	32,500
4	1,250	1,200	1,000	1,550	2,500	7,500
5	450	400	350	500	650	2,350
Total	106,700	99,100	85,850	127,550	225,650	644,850

SDG 8 Decent Work and Economic Growth

⁴¹ Refer to the registered GS project Aqua Clara Water Filtration Program in Kenya (GS 1078). <https://registry.goldstandard.org/projects/details/59>

As described in section B.6.1 of this PDD, the number of jobs created for males and females will be recorded. Percentage of employees with salaries paid at par with the average wage of Kenya can be calculated as:

The number of employees that are paid at par with the average wage of Kenya/The number of jobs created*100%.

The outcomes for SDG 8 are shown as the following table.

Year	Number of males and females employed	Percentage of employees with salaries paid more than the minimum wage of Kenya
1	12 people (6 males, 6 females)	100%
2	12 people (6 males, 6 females)	100%
3	12 people (6 males, 6 females)	100%
4	12 people (6 males, 6 females)	100%
5	12 people (6 males, 6 females)	100%
Total	12 people (6 males, 6 females)	100%

SDG 13 Climate Action

The detailed calculation method is shown in section B.6.1 of this PDD. The transparent ex-ante calculations of the outcome for SDG 13 (GHG emission reductions) are provided in a separate Excel Spreadsheet uploaded to GS Registry. For data/parameters available at the time of design certification, values contained in section B.6.2 of this PDD and for data/parameters only available after monitoring the estimates contained in section B.7.1 of this PDD have been used. The outcomes for SDG 13 are shown as the following table.

Year	Emission reductions (tCO _{2e})					
	Makueni	Trans Nzoia	Kirinyaga	Bungoma	Nakuru	Total
1	76,061	62,389	64,855	78,424	86,340	368,069
2	95,076	77,362	80,321	98,236	109,888	460,883
3	99,830	81,105	84,312	103,602	116,429	485,278
4	101,018	82,104	85,310	104,882	117,737	491,050
5	101,446	82,436	85,659	105,295	118,077	492,913
Total	473,431	385,396	400,456	490,439	548,471	2,298,193

B.6.2. Summary of ex ante estimates of each SDG Impact

Ex-ante estimation of SDG 6 Outcome:

Year	Baseline estimate	Project estimate	Net benefit
1	0 person.days	553,423,842 person.days	553,423,842 person.days
2	0 person.days	694,662,218 person.days	694,662,218 person.days
3	0 person.days	732,133,624 person.days	732,133,624 person.days
4	0 person.days	740,780,871 person.days	740,780,871 person.days
5	0 person.days	743,490,342 person.days	743,490,342 person.days
Total	0 person.days	3,464,490,896 person.days	3,464,490,896 person.days
Total number of crediting years	5		
Annual average over the crediting period	0 person.days	692,898,179 person.days	692,898,179 person.days

Ex-ante estimation of SDG 7 Outcome:

Year	Baseline estimate	Project estimate	Net benefit
1	0 water purifier	480,000 water purifiers	480,000 water purifiers
2	0 water purifier	122,500 water purifiers	122,500 water purifiers
3	0 water purifier	32,500 water purifiers	32,500 water purifiers
4	0 water purifier	7,500 water purifiers	7,500 water purifiers
5	0 water purifier	2,350 water purifiers	2,350 water purifiers
Total	0 water purifier	644,850 water purifiers	644,850 water purifiers
Total number of crediting years	5		
Annual average over the crediting period	0 water purifier	128,970 water purifiers	128,970 water purifiers

Ex-ante estimation of SDG 8 Outcome:

Year	Baseline estimate	Project estimate	Net benefit
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1	0 jobs created, 0% of employees with salaries paid more than the minimum wage of Kenya	12 jobs created including 6 males and 6 females, 100% of employees with salaries paid more than the minimum wage of Kenya	12 jobs created including 6 males and 6 females, 100% of employees with salaries paid more than the minimum wage of Kenya
2	0 jobs created, 0% of employees with salaries paid more than the minimum wage of Kenya	12 jobs created including 6 males and 6 females, 100% of employees with salaries paid more than the minimum wage of Kenya	12 jobs created including 6 males and 6 females, 100% of employees with salaries paid more than the minimum wage of Kenya
3	0 jobs created, 0% of employees with salaries paid more than the minimum wage of Kenya	12 jobs created including 6 males and 6 females, 100% of employees with salaries paid more than the minimum wage of Kenya	12 jobs created including 6 males and 6 females, 100% of employees with salaries paid more than the minimum wage of Kenya
4	0 jobs created, 0% of employees with salaries paid more than the minimum wage of Kenya	12 jobs created including 6 males and 6 females, 100% of employees with salaries paid more than the minimum wage of Kenya	12 jobs created including 6 males and 6 females, 100% of employees with salaries paid more than the minimum wage of Kenya
5	0 jobs created, 0% of employees with salaries paid more than the minimum wage of Kenya	12 jobs created including 6 males and 6 females, 100% of employees with salaries paid more than the minimum wage of Kenya	12 jobs created including 6 males and 6 females, 100% of employees with salaries paid more than the minimum wage of Kenya
Total	0 jobs created, 0% of employees with salaries paid more than the minimum wage of Kenya	12 jobs created including 6 males and 6 females, 100% of employees with salaries paid more than the minimum wage of Kenya	12 jobs created including 6 males and 6 females, 100% of employees with salaries paid more than the minimum wage of Kenya

Total number of crediting years	5		
Annual average over the crediting period	0 jobs created, 0% of employees with salaries paid more than the minimum wage of Kenya	12 jobs created including 6 males and 6 females, 100% of employees with salaries paid more than the minimum wage of Kenya	12 jobs created including 6 males and 6 females, 100% of employees with salaries paid more than the minimum wage of Kenya

Ex-ante estimation of SDG 13 Outcome:

Year	Baseline estimate	Project estimate	Net benefit
1	368,069 tCO ₂ e	0 tCO ₂ e	368,069 tCO ₂ e
2	460,883 tCO ₂ e	0 tCO ₂ e	460,883 tCO ₂ e
3	485,278 tCO ₂ e	0 tCO ₂ e	485,278 tCO ₂ e
4	491,050 tCO ₂ e	0 tCO ₂ e	491,050 tCO ₂ e
5	492,913 tCO ₂ e	0 tCO ₂ e	492,913 tCO ₂ e
Total	2,298,193 tCO₂e	0 tCO₂e	2,298,193 tCO₂e
Total number of crediting years	5		
Annual average over the crediting period	459,638 tCO ₂ e	0 tCO ₂ e	459,638 tCO ₂ e

B.7. Monitoring plan

B.7.1 Data and parameters to be monitored

Related to water quality

Parameter ID	SDWS 20
Data / Parameter	Water hygiene education campaigns
Unit	N/A
Description	Hygiene campaigns carried out among project technology users. The drinking water and hygiene practices will be assessed using the WHO/UNICEF Joint Monitoring Programme guidelines ⁴² (JMP 2018). In-person or telephone or by messaging based survey will be conducted including all the JMP core questions for drinking water and core questions for hygiene. Refer to section B.7.3 of the PDD for more details.
Source of data	Hygiene practices assessment

⁴² <https://washdata.org/report/jmp-2018-core-questions-household-surveys>

Value(s) applied	N/A
Measurement methods and procedures	The water hygiene education campaigns will be conducted on the users and the JMP questionnaire which including all the JMP core questions for drinking water and core questions for hygiene will be supplied.
Monitoring frequency	Annually
QA/QC procedures	The fraction of the households where Safe water and Hygiene practices are found to fulfill “safely managed” or “basic” requirements is expected to increase over time as a result of the hygiene campaigns. The monitoring indicators for this project are defined as: The fraction of the households where safe water is found to fulfill "safely managed" is defined as the fraction of the households involved in this project who can obtain safe water. The fraction of the households where Hygiene practices are found to fulfill “basic” requirements can be defined as the fraction of the soap availability for handwashing.
Purpose of data	
Additional comment	N/A

SDG 7

Parameter ID	SDWS 19																																																							
Data / Parameter	Number of water purifiers sold in year y																																																							
Unit	Sets																																																							
Description	Number of water purifiers sold in year y																																																							
Source of data	Sales database																																																							
Value(s) applied	Will be reported for each monitoring period. According to the sales plan during the first crediting period, the estimated number of water purifiers sold in year y are shown as the following table. <table><tr><th rowspan="2">Year</th><th colspan="6">Estimated number of water purifiers sold in year y (sets)</th></tr><tr><th>Makueni</th><th>Trans Nzoia</th><th>Kirinyaga</th><th>Bungoma</th><th>Nakuru</th><th>Total</th></tr><tr><td>1</td><td>80,000</td><td>75,000</td><td>65,000</td><td>95,000</td><td>165,000</td><td>480,000</td></tr><tr><td>2</td><td>20,000</td><td>18,000</td><td>15,500</td><td>24,000</td><td>45,000</td><td>122,500</td></tr><tr><td>3</td><td>5,000</td><td>4,500</td><td>4,000</td><td>6,500</td><td>12,500</td><td>32,500</td></tr><tr><td>4</td><td>1,250</td><td>1,200</td><td>1,000</td><td>1,550</td><td>2,500</td><td>7,500</td></tr><tr><td>5</td><td>450</td><td>400</td><td>350</td><td>500</td><td>650</td><td>2,350</td></tr><tr><td>Total</td><td>106,700</td><td>99,100</td><td>85,850</td><td>127,550</td><td>225,650</td><td>644,850</td></tr></table>	Year	Estimated number of water purifiers sold in year y (sets)						Makueni	Trans Nzoia	Kirinyaga	Bungoma	Nakuru	Total	1	80,000	75,000	65,000	95,000	165,000	480,000	2	20,000	18,000	15,500	24,000	45,000	122,500	3	5,000	4,500	4,000	6,500	12,500	32,500	4	1,250	1,200	1,000	1,550	2,500	7,500	5	450	400	350	500	650	2,350	Total	106,700	99,100	85,850	127,550	225,650	644,850
Year	Estimated number of water purifiers sold in year y (sets)																																																							
	Makueni	Trans Nzoia	Kirinyaga	Bungoma	Nakuru	Total																																																		
1	80,000	75,000	65,000	95,000	165,000	480,000																																																		
2	20,000	18,000	15,500	24,000	45,000	122,500																																																		
3	5,000	4,500	4,000	6,500	12,500	32,500																																																		
4	1,250	1,200	1,000	1,550	2,500	7,500																																																		
5	450	400	350	500	650	2,350																																																		
Total	106,700	99,100	85,850	127,550	225,650	644,850																																																		
Measurement methods and procedures	The number of water purifiers sold will be based on the sale database and cross checked by the purchase receipt.																																																							
Monitoring frequency	Yearly																																																							
QA/QC procedures	The number of water purifiers sold will be based on the sale database and cross checked by the purchase receipt.																																																							
Purpose of data	To demonstrate contribution to SDG 7.																																																							
Additional comment	N/A																																																							

SDG 6/SDG 13

Parameter ID	SDWS 18
Data / Parameter	$M_{q,y}$
Unit	%
Description	Ongoing water quality indicated as the fraction of the samples that pass microbial quality standard requirements specified in relevant microbial quality standard for drinking water of the host country (<i>KS EAS 153: 2018 East African Standard Specification for Packaged Drinking Water</i>).
Source of data	For HWT, testing of the water that exits the treatment technology.
Value(s) applied	Will be reported for each monitoring period. For ex ante estimation, 100% is taken based on the manufacturer's specifications of the project technologies.
Measurement methods and procedures	The water quality test applies the bacterial quality standard <1 cfu/100ml, and the sampling determines the proportion of pass and fail results. The sample size is following 90/10 precision rule and a minimum sample size of 30 must be selected. The water sample is taken at water outlet of the project water purifiers randomly from project database.
Monitoring frequency	Annual sampling, and the first round of testing shall be conducted at least after six months from the start date.
QA/QC procedures	1. Laboratories used for water quality testing must be approved by local health authorities and/or have quality accreditation; and 2. The laboratory used must demonstrate that it has an adequate quality management plan in place which addresses both quality assurance and quality control test procedures. 3. Field testing kits also are eligible, e.g. based on Colony Forming Unit method or Most Probable Number method. To use the field testing kits the project shall meet the following requirements: a. Testing kits must be approved by national agency or meet standards set by relevant international organization e.g., US-EPA, and b. Testing kits shall be tested for its accuracy and robustness prior to application for project level monitoring, whereby local or accredited laboratory shall conduct water quality tests using testing kits and a relevant ISO standard or an equivalent standard, in parallel with field testing kits.
Purpose of data	To demonstrate contribution to SDG 6 and SDG 13.
Additional comment	If the proportion of samples not meeting Safe Drinking Water Quality Standards exceeds a threshold, no emission reductions can be claimed for the corresponding monitoring period. Thresholds: - Project or VPA year 1: 20%

	- Project or VPA year 2: 15% - Project or VPA year 3 or above: 10% When the crediting period is renewed, the year number count continues, i.e., the second crediting period would encompass year 6, year 7, year 8, etc. Additionally, when the threshold is exceeded, the project shall provide an explanation for why this occurred and provide a remediation plan.
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Parameter ID	SDWS 25
Data/parameter	$HN_{p,y}$
Unit	Number
Description	Number of individuals per premises type p in the project boundary in year y
Source of data	Project Survey which will be conducted annually in the monitoring period.
Value(s) applied	3.9
Measurement methods and procedures	More than 100 households will participate in the project survey. Refer to section B.7.2 and B.7.3 of the PDD for more details. For ex ante estimation, the results from baseline survey will be used. 505 households participated in the baseline survey, which included 1,994 individuals. $HN_{p,y}$ can be calculated as $1,994/505=3.9$.
Monitoring frequency	Annual
QA/QC procedures	The value applied shall be cross-checked against at least one other source on the list. For cross-check purposes, sources applied may be up to 5 years old. Further, cross-check with older sources may be used provided they provide conservative results.
Purpose of data	To demonstrate contribution to SDG 13.
Additional comment	As per <i>2019 Kenya Population and Housing Census Volume I: Population by County and Sub-County</i> ⁴³ , there are 1,647,507 households and 6,421,177 people in Makueni County, Trans Nzoia County, Kirinyaga County, Bungoma County and Nakuru County of Kenya. Number of individuals per household can be calculated as $6,421,177/1,647,507=3.9$, which is consistent with the baseline survey result.

Parameter ID	SDWS 28
Data / Parameter	$N_{p,y}$
Unit	Number

⁴³ <https://www.knbs.or.ke/?wpdmpo=2019-kenya-population-and-housing-census-volume-i-population-by-county-and-sub-county>

Description	Accumulated number of premises type p with at least one individual project technology in year y																		
Source of data	Sales database																		
Value(s) applied	Will be reported for each monitoring period. In order to cover more households with high-quality water purifiers at a lower price, it is restricted in the project that each household can only buy one set of water purifier. According to the sales plan during the first crediting period, the results of the parameter are: <table><tr><th>Year</th><th>The water purifiers sold in year y (Sets)</th><th>$N_{p,y}$ (Sets)</th></tr><tr><td>1</td><td>480,000</td><td>480,000</td></tr><tr><td>2</td><td>122,500</td><td>602,500</td></tr><tr><td>3</td><td>32,500</td><td>635,000</td></tr><tr><td>4</td><td>7,500</td><td>642,500</td></tr><tr><td>5</td><td>2,350</td><td>644,850</td></tr></table>	Year	The water purifiers sold in year y (Sets)	$N_{p,y}$ (Sets)	1	480,000	480,000	2	122,500	602,500	3	32,500	635,000	4	7,500	642,500	5	2,350	644,850
Year	The water purifiers sold in year y (Sets)	$N_{p,y}$ (Sets)																	
1	480,000	480,000																	
2	122,500	602,500																	
3	32,500	635,000																	
4	7,500	642,500																	
5	2,350	644,850																	
Measurement methods and procedures	Sale database will be recorded and maintained according to the monitoring plan. In addition, the sale database will contain the following contents: i. Date of sale ii. Geographic area of sale iii. Model/type of project technology sold iv. Quantity of project technology sold Name and telephone number, and address (if available) or other traceable indicator of premises identity and location for all end users.																		
Monitoring frequency	Annually																		
QA/QC procedures	The number of water purifiers sold will be cross checked by the purchase receipt.																		
Purpose of data	To demonstrate contribution to SDG 6 and SDG 13.																		
Additional comment	Units shall not be counted in $N_{p,y}$ after the end of their technical life, unless this is addressed by the measures to manage the cases where the expected technical life of the project technology is shorter than the crediting period.																		

Parameter ID	SDWS 31
Data / Parameter	$DP_{p,y}$
Unit	Days
Description	Average days the project technology is present for end-users in the premises p in year y
Source of data	Sales database
Value(s) applied	For ex ante estimation, 365 days will be taken.
Measurement methods and procedures	Based on the records of "Date of sale" and ex-ante parameter "Expected technical life of project technology," determine for

	each project device how many days of the 365 days of the year it was in a household and within its technical life. Calculate the average for all the project devices to obtain this parameter.
Monitoring frequency	Annually
QA/QC procedures	N/A
Purpose of data	To demonstrate contribution to SDG 6 and SDG 13.
Additional comment	N/A

Parameter ID	SDWS 29
Data / Parameter	$U_{p,y}$
Unit	%
Description	Usage rate of the project technology by premises type p during year y
Source of data	Annex 1: Usage Surveys Guidelines – HWT Technologies in the applied methodology: METHODOLOGY FOR EMISSION REDUCTIONS FROM SAFE DRINKING WATER SUPPLY (version 1.0)
Value(s) applied	Will be reported for each monitoring period. For ex ante estimation, 80% is used as per the registered GS project Aqua Clara Water Filtration Program in Kenya (GS 1078) ⁴⁴ .
Measurement methods and procedures	In-person survey of project households following Annex 1 of the applied GS methodology. Households that show at least once-in-two-days use may be counted as users. The resulting fraction is multiplied by 100% to get $U_{p,y}$. The minimum total sample size is 100, with at least 30 samples for project technologies of each age being credited. Where project technologies of different ages are being credited, the sample shall be representative of the distribution of project technology ages.
Monitoring frequency	Annually
QA/QC procedures	Water Consumption Field Test (WCFT) may be used to cross check the usage percentage
Purpose of data	To demonstrate contribution to SDG 6 and SDG 13.
Additional comment	The usage survey provides a single usage parameter that is representative for project technologies in the total sales record.

SDG 13

Parameter ID	SDWS 24
Data / Parameter	QPW_p
Unit	Liters/person/day

⁴⁴ <https://registry.goldstandard.org/projects/details/59>

Description	Volume of drinking water per person per day for premises type p
Source of data	Water Consumption Field Test (WCFT)
Value(s) applied	Will be reported for each monitoring period. For ex ante estimation, the result of 4.3 L/person/day from baseline survey will be used.
Measurement methods and procedures	The WCFT is conducted with end user's representative of the project scenario target population and currently using the project technology. The WCFT must be designed to ensure that monitoring is representative of typical technology use practices and that: - it is transparent and can easily be replicated, - it is evidently conservative, - the sample is randomly selected so as to not introduce a material bias, and - the impact of daily and seasonal variations on the expected average water consumption is accounted for The WCFT must be conducted over 3 days, not including weekends, and averaged value (L/person/day) value should be determined after excluding outliers. It must be made explicit to the households that they must behave and consume water normally, reflecting typical daily water consumption pattern. Minimum sample size is 30. Any sampling methods can be used, provided that the sample is selected randomly. Refer to section B.7.2 and B.7.3 of the PDD for more details.
Monitoring frequency	Every two years
QA/QC procedures	-
Purpose of data	To demonstrate contribution to SDG 13.
Additional comment	The value is capped at 5.5 L/person/day, which is determined based on WHO recommendations (Domestic Water Quantity, Service Level and Health, Table 2: Volumes of water required for hydration, WHO 2003). https://www.who.int/water_sanitation_health/diseases/WSH03.02.pdf.

Parameter ID	SDWS 32
Data / Parameter	$DN_{p,y}$
Unit	Number
Description	Average number of individual project technologies in each project premises type p in year y
Source of data	Sales database

Value(s) applied	In order to cover more households with high-quality water purifiers at a lower price, it is restricted in the project that each household can only buy one set of water purifier.
Measurement methods and procedures	Based on the sales records of "Quantity of project technologies sold" and identifying information of buyer/recipient, calculate the average number of project devices per household.
Monitoring frequency	Annually
QA/QC procedures	-
Purpose of data	To demonstrate contribution to SDG 13.
Additional comment	N/A

Parameter ID	SDWS 30
Data / Parameter	$t_{p,y}$
Unit	Hours per day
Description	Usage time of the project technology by premises type p in year y
Source of data	<i>Methodology for Emission Reductions from Safe Drinking Water Supply</i> (Version 1.0)
Value(s) applied	5 hours.
Measurement methods and procedures	Option 3. Default of 5 hours.
Monitoring frequency	Annually
QA/QC procedures	-
Purpose of data	To demonstrate contribution to SDG 13.
Additional comment	N/A

Parameter ID	SDWS 22
Data / Parameter	$X_{cleanboil,y}$
Unit	%
Description	Proportion of project households that boil safe (treated, or from safe supply) water after installation of project technology in year y
Source of data	Project survey
Value(s) applied	Will be reported for each monitoring period. For ex ante estimation, 0% from baseline survey will be taken.
Measurement methods and procedures	In-person observational sample-based survey of project household practices. The safe water project survey has the same requirements as the baseline survey (see Table B4.1), but it is conducted with end user's representative of the project scenario target population and currently using the safe water project technology. Guidance on representativeness and sample sizing is the same.

Monitoring frequency	Annually
QA/QC procedures	-
Purpose of data	To demonstrate contribution to SDG 13.
Additional comment	N/A

Parameter ID	SDWS 35
Data / Parameter	LE_y
Unit	tCO ₂ e per year
Description	Leakage emissions during year y
Source of data	Sources established by above Leakage emissions section
Value(s) applied	0 tCO ₂ e/y and will be assessed every two years.
Measurement methods and procedures	As per section 3.8 of the applied GS methodology.
Monitoring frequency	Every two years.
QA/QC procedures	Compliance with the general requirements for sampling and general requirements for data and information sources
Purpose of data	To demonstrate contribution to SDG 13.
Additional comment	N/A

SDG 8

Parameter ID	SDWS 19
Data / Parameter	<i>Number of males and females employed</i>
Unit	Number
Description	Jobs created for both men and women.
Source of data	Employment records
Value(s) applied	Will be reported for each monitoring period. For ex ante estimation, 12 jobs are created including 6 males and 6 females.
Measurement methods and procedures	The number of persons employed will be calculated based on the employment records and cross checked by the labor contracts.
Monitoring frequency	Once for each monitoring period.
QA/QC procedures	Data will be collected using the standard procedures and will be stored for the crediting period and an additional two years after the end of the crediting period or the last issuance of GS VERs for this project activity, whichever occurs later.
Purpose of data	To demonstrate contribution to SDG 8.
Additional comment	N/A

Parameter ID	SDWS 19
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Data / Parameter	<i>Percentage of employees with salaries paid at par with the average wage of Kenya.</i>
Unit	%
Description	This parameter will be collected to indicate the project provides decent work for both men and women.
Source of data	Employment Payroll and average wages in Kenya based on most recent information available.
Value(s) applied	Will be reported for each monitoring period. Kenya's Average Wage Earnings data was reported at 778,248 KES in 2019. The project owner will guarantee that the average wage of the employees will be higher than that of Kenya in 2019 ⁴⁵ . Therefore, for ex ante estimation, 100% is taken.
Measurement methods and procedures	The number of employees that are paid at par with the average wage of Kenya/The number of jobs created*100%.
Monitoring frequency	Once for each monitoring period.
QA/QC procedures	Data will be collected using the standard procedures and will be stored for the crediting period and an additional two years after the end of the crediting period or the last issuance of GS VERs for this project activity, whichever occurs later.
Purpose of data	To demonstrate contribution to SDG 8.
Additional comment	N/A

B.7.2 Sampling plan

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The data source of parameters to be monitored are summarized as follows, in which sampling approach is not applicable for the parameters that are source from sales database, employment records, default value and leakage assessment:

- Sales database: *Number of water purifiers sold in year y , $N_{p,y}$* (Accumulated number of premises type p with at least one individual project technology in year y), $DP_{p,y}$ (Average days the project technology is present for end-users in the premises p in year y) and $DN_{p,y}$ (Average number of individual project technologies in each project premises type p in year y).
- Employment records: *Number of males and females employed*, and *Percentage of employees with salaries paid at par with the average wage of Kenya*.
- Default value: $t_{p,y}$ (Usage time of the project technology by premises type p in year y).
- Leakage assessment: LE_y (Leakage emissions during year y).

The parameters that need sampling to monitoring are:

⁴⁵ <https://www.ceicdata.com/en/kenya/average-wage-earnings-by-sector-and-industry-international-standard-of-industrial-classification-rev-4/average-wage-earnings>

- e. Project survey: $X_{cleanboil,y}$ (Proportion of project end-users that boil safe (treated, or from safe supply) water after installation of project technology in year y) and $HN_{p,y}$ (Number of individuals per premises type p in the project boundary in year y).
- f. Water Consumption Field Test (WCFT): QPW_p (Volume of drinking water per person per day for premises type p).
- g. Usage survey: $U_{p,y}$ (Usage rate of the project technology by premises type p during year y).
- h. Water quality survey test: $M_{q,y}$ (Ongoing water quality indicated as the fraction of the samples that pass the test of bacterial quality to a standard of <1cfu/100ml).
- i. Water hygiene education campaigns.

The sampling approach is to be used for following surveys and tests as summarized below:

B.7.2.1 Project Survey

The safe water project survey is conducted with end user's representative of the project scenario target population and currently using the safe water project technology. The project survey will be carried out using representative and random sampling following the GS guidelines for minimum sample size:

Group size <300: Minimum sample size 30

Group 300 to minimum to 1,000: Minimum sample size 10% of group size

Group size >1,000: Minimum sample size 100

The project survey will likely have a minimum sample size of 100 as the number of water purifier sold is estimated to be greater than 1,000.

End users for the project survey will be selected using representative sampling techniques to ensure adequate representation of users with technologies of different ages. Common sampling approach such as clustered random sampling is allowed, and geographic distribution will be factored into selection criteria⁴⁶. End users can be surveyed at any time throughout the year with care taken to collect information pertaining to seasonal variations in technology and fuel use patterns.

B.7.2.2 WCFT

For WCFT statistical analysis, 90/10 precision level will be selected. The survey will be a single sample for establishing project scenario water consumption. The calculation of the sample size

⁴⁶ CDM Guideline: *Sampling and surveys for CDM project activities and programmes of activities* (Version 04.0). https://cdm.unfccc.int/filestorage/e/x/t/extfile-20151023152925164-Meth_GC48_-_ver04.0-.pdf/Meth_GC48_%28ver04.0%29?t=ZGd8cWp4ZnozFDAb8zgNROkmMV4uQ-CRDO0L.

for the survey shall be based upon a 0.6 Coefficient of Variation (COV) under a 90/10 approach as provided in Table 3 of Annex 4 of *TPDDTEC* (Version 3.1).

Table 3 of Annex 4 of *TPDDTEC* (Version 3.1): Sample sizes in cases of SINGLE samples (where the tests are conducted for either baseline or project scenario but not both).

COV	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
90/10 precision	12	26	45	70	101	137	179	226	279

Table 3 is applicable because the survey will be conducted for the project scenario as a single sample. The selection of a COV of 0.6 was arbitrarily determined from the project proponent's assumption which is allowed in Annex 4 of *TPDDTEC* (Version 3.1) "Use the tables here to choose a provisional minimum sample size using that COV estimate." Furthermore, the PP attests herein that the COV may be increased if after analysis of the data and results presented, the COV used (0.6) is higher than the real COV after analysis of the data collated from the monitoring exercise.

The Project Survey and the WCFT can be conducted concurrently using the same sample. The sample size for project survey is 100, the sample size for WCFT is 101, so the sample size for together conducted of Project Survey and the WCFT is 101.

B.7.2.3 Usage Survey

The minimum total sample size for Usage Survey is 100, with at least 30 samples for project technologies of each age being credited⁴⁷. The majority of interviews in a usage survey must be conducted in person. Thus, if technologies of age 1-5 are credited, the usage survey must include 30 representative samples from each age for total of 150 samples. The resulting usage parameter should be weighted based on the proportion of technologies in the borehole database of each age.

In line with the applied GS methodology, any sampling methods can be used, provided that the sample is selected randomly. If sampling approach other than simple random sampling is applied, CDM Guideline: *Sampling and surveys for CDM project activities and programmes of activities* (Version 04.0) must be followed.

B.7.2.4 Water Quality Testing

⁴⁷ Age of the water purifiers will be determined based on the sales database.

The sample size is following 90/10 precision rule and the water sample is taken at water outlet randomly from project database. According to the applied methodology, a minimum sample size of 30 must be selected.

Once the sample users determined, the monitoring of this parameter can be conducted, the water sample should be taken from the outlet of samples water purifier and the water sample quality test should be Entrust a qualified third-party professional institution to conducted. For this project, the third-party professional test institution is KEBS⁴⁸ (Kenya Bureau of Standards), which is official organization with the ability to issue qualifications to other institution.so, the KEBS has the undoubted accreditation.

B.7.2.5 Water Hygiene Education Campaigns

The calculation of the sample size for the test shall be based upon a 0.6 Coefficient of Variation (COV) under a 90/10 approach as provided in Table 3 of Annex 4 of *TPDDTEC* (Version 3.1).

Table 3 is applicable because the survey will be conducted for the project scenario as a single sample. The selection of a COV of 0.6 was arbitrarily determined from the project proponent's assumption which is allowed in Annex 4 of *TPDDTEC* (Version 3.1) "Use the tables here to choose a provisional minimum sample size using that COV estimate." Furthermore, the project proponent attests herein that the COV may be increased if after analysis of the data and results presented, the COV used (0.6) is higher than the real COV after analysis of the data collated from the monitoring exercise.

The Water Quality Testing and the Water Hygiene Education Campaigns can be conducted concurrently using the same sample. As per applied methodology, the sample size for Water Quality Testing has the minimum size of 30, the sample size for Water Hygiene Education Campaigns is 101, so the sample size for together conducted of Water Quality Testing and the Water Hygiene Education Campaigns is 101. The sample size for Water Quality Testing has the minimum size of 30, the actual sample size is 101 which higher 30. As the larger the sample size, the lower the sampling risk and the more accurate the estimate, so the 101 sample can be used for the monitoring of the water quality testing and the Water Hygiene Education Campaigns.

The WQT and Hygiene survey use the same sample of users. The procedure of WQT is that the project monitoring staff lead the laboratory sampling personnel to the sample user's home to take a water sample (including raw water and filtered water), and then the water sample is

⁴⁸ https://www.kebs.org/index.php?option=com_content&view=article&id=94:microbiology-laboratory&catid=30&Itemid=161

appropriately stored and then brought back to the laboratory for testing. Hygiene survey is to monitor employees to conduct surveys at users' homes and fill out questionnaires. Therefore, when the samples are determined, the project monitoring staff will lead the laboratory sampler to the sample user's home, and then the sampler can collect water samples, and the project monitoring staff will conduct a questionnaire survey on the main users of the water purifier. Both can be done at the same time.

B.7.3 Other elements of monitoring plan

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The overall monitoring structure of the project shows as below. The project owner and local participants will establish GS Monitoring Team for the target areas to be monitored within the project boundary to collect and record monitoring data. The GS Monitoring Team includes three groups, Sales Group, Survey Group and Record Group. The candidates will be carefully screened to ensure that each GS group member has the proper ability to perform the monitoring task.

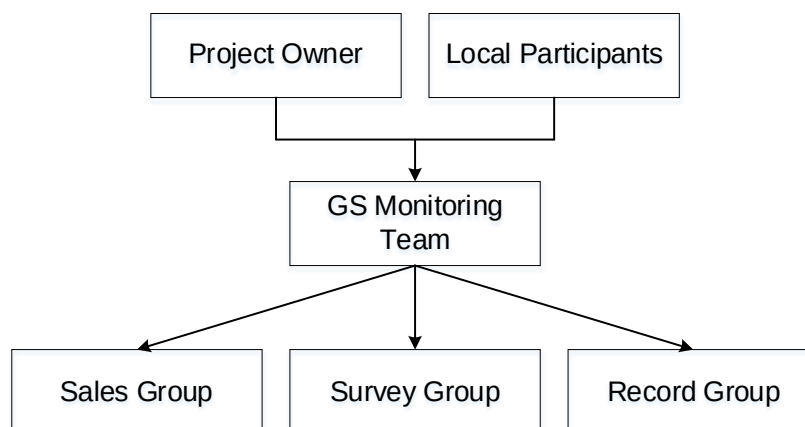


Figure B7.1 The organization of the project monitoring

Responsibility of each group are summarized as follows:

Local participants: For this project, the local participant is TRIOPT AFRICA LIMITED⁴⁹, which is a subsidiary of TAL HOLDING LIMITED (one of the funders of the project owner CNKL). Based on the trend of common interests and trust, a better internal communication mechanism will be more conducive to the development of sales and develop the project as an emission reduction project under the GS mechanism.

TRIOPT AFRICA LIMITED is a subsidiary under TAL HOLDING LIMITED, which is an independent, ambitious and fast growing in African telecom sector. It has years of professional

⁴⁹ TRIOPT AFRICA LIMITED is a subsidiary of TAL HOLDING LIMITED, the project owner CNKL is established by TAL HOLDING LIMITED cooperated with ROTOR ENERGY TECHNOLOGY CORPORATION LIMITED, so the TRIOPT AFRICA LIMITED and the project owner CNKL has the same investor.

distribution experiences in telecom industry within East Africa across Kenya, Tanzania, Uganda, Rwanda and Mozambique. TAL HOLDING LIMITED had cooperated with ROTOR ENERGY TECHNOLOGY CORPORATION LIMITED to carry out Akon Lighting Africa Initiative, which covered 25 African counties, provided solutions to the energy poverty in rural areas in Africa through innovative, clean and accessible solar solutions. The shareholding structure is as follows:

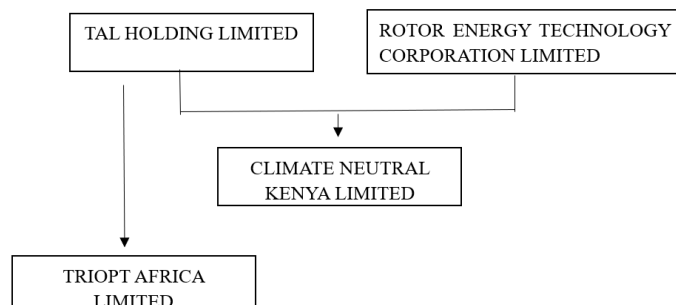


Figure B7.2 The shareholding structure of the project owner and local participant

For this project, Climate Neutral Kenya Limited took advantage of the sales network that TRIOPT AFRICA LIMITED had built and delegated the sales work to TRIOPT AFRICA LIMITED, with their help, achieved gratifying sales performance. The specific sales model of this project is as follows: In areas with relatively prosperous economy, such as main urban areas and counties, TRIOPT has set up special sales points to sell products directly. In relatively remote townships and rural areas where the economy is slightly underdeveloped, the sales work was conducted by using the distribution network (for example, township convenient stores, rural shops) established by TRIOPT, i.e., using the local resellers developed by TRIOPT to sell the water purifiers. At the same time, each county will establish a sales management center to be responsible for statistics and analysis of sales in each county, and to solve various problems from the sales process.

No matter what kind of sales model, at a time of water purifier pick up, the end user signs a sales contract with the project owner to give up carbon emission ownership and get a receipt issued by the seller. The contract and a copy of this receipt are kept by project owner. The sales management center picks up the sales contract weekly.

In addition, when the product is sold, the salesperson will record the buyer's information (such as name, gender, age, address etc.) and the water purifier serial number. The project record adopts a sales management system, which can perform real-time numbering according to the order of sales of water purifiers and can generate the excel sheet of Sales Record Database.

The GS monitoring team used the sales contract to cross-checked with the Sales Record Database.

Sales group: Maintain and updated the sales database in a timely manner. The project proponent must maintain an accurate and complete electronic database for enabling the unique identification of the project water purifiers. It is envisaged that the database will include at least the following information:

- Date of sale
- Geographic area of sale
- Model/type of project technology sold and unique water purifier serial number
- Mode of use: commercial, institutional, domestic and other
- Number of water purifiers sold
- Name, telephone number and address.

Survey group: To carry out project survey, usage survey, water quality survey test and hygiene practices assessment.

Record group: Data verification and filing and managing the project documents.

As per the applied GS methodology, the project proponent must conduct the following studies for each project scenario:

B.7.3.1 Project Non-Renewable Biomass (NRB) Assessment - at renewal of crediting period

In accordance with the applied GS methodology, non-renewable biomass fraction is fixed for the first crediting period based on the results of the *NRB* assessment. In case of a renewal of the crediting period and as per Gold Standard rules, the *NRB* fraction must be reassessed as any other baseline parameters and updated in line with most recent data available.

B.7.3.2 Project Survey - completed annually

The project survey investigates changes over time in a project scenario. It provides critical information on year-to-year trends in end user characteristics such as technology use, fuel consumption and seasonal variations. Questionnaire which is similar to the baseline survey (Refer to Table B4.1 of this PDD for sample questionnaire) will be adapted. It will be conducted with end-users representative of the project scenario target population and currently using the safe water project technology. Related data and information will be collected through in-person paper surveys, which will be then entered into an excel file to be used to analyze the data.

In the guidance on data collected, questions about end user characteristics and baseline technology and fuels should be treated as specific to safe water supply and boiling. These questions should be asked twice, first in regard to the baseline scenario water supply and water

treatment, including boiling technologies, and second in regards to the project scenario clean water supply, including treatment and boiling technologies.

B.7.3.3 Water Consumption Field Test (WCFT) - completed every two years

The water consumption field test (WCFT) measures project-supplied clean water consumption volumes. The WCFT is conducted with end user's representative of the project scenario target population and currently using the project technology. The WCFT must be designed to ensure that monitoring is representative of typical technology use practices and that:

- it is transparent and can easily be replicated,
- it is evidently conservative,
- the sample is randomly selected so as to not introduce a material bias, and
- the impact of daily and seasonal variations on the expected average water consumption is accounted for.

The WCFT must be conducted over 3 days, not including weekends, and averaged value (L/person/day) value should be determined after excluding outliers. It must be made explicit to the households/institutions that they must behave and consume water normally, reflecting typical daily water consumption pattern. WCFT protocol are designed for the project based on the Kitchen Project Test (KPT) guidelines in Annex 4 of *TPDDTEC* (Version 3.1). The protocol will be executed over a four-day period with the first day targeted at questionnaires and surveys with the remaining three left dedicated to WCFT exercises. The four-day period shall not be carried out over a holiday period to ensure conservative quantification of water consumption per end user.

Day 0: Training of field enumerators

The PP will carry out training exercise with the enumerators who would then proceed to carry out the WCFT. The PP will go over the survey questions to be asked during WCFT exercise as well as did mock examples of real WCFT with the enumerators to ensure they understood what would take place in a real-world situation in the field. Calibrated buckets are issued to each of the enumerators which they would use in the field in measuring the water volume consumed in a day per household. Enumerators are expressly told to ensure maximum accuracy in measuring water using the buckets.

Day 1 - Surveys and Introduction

The project proponent will inform the end-users of the 3-day WCFT exercise that them that they should treat water as usual. Households will report on the times at which they treat water every day. The times reported shall be recorded to allow enumerators to go to the end-user's residence

the following day before raw water is treated. Distribute calibration flasks for measurement to users.

Day 2 - 4. WCFT exercise

Distribute a Water consumption survey record form to users every morning before the user's beginning to use water. When users want to use water, use a volumetric flask to collect water from the water purifier and record the amount of water received. Daily Water consumption by the end users will be measured in these flasks filtered through the water filter. In every evening the Water consumption survey record form will be collected.

The mean value of the total water consumption (the total water consumption per day aggregated over a three-day period) will establish the value QPW_p per end user. Which will then be aggregated (after elimination of outliers) to determine the value used to calculate Emission Reductions for the whole project per technology.

The end user will then be asked if at all they boil any of the clean water or if they consume any raw water without boiling.

Water consumption for the water filter will be capped at 5.5 litres per person per day in accordance with the applied GS methodology.

B.7.3.4 On-going Monitoring Studies: Usage rates, leakage, water quality

Usage Survey - completed annually

In-person survey of project households will be conducted following Annex 1 of the applied GS methodology. The guidelines outline six topics which must be successfully completed for the survey respondent to be classed as a user. Additionally, other topics relevant to specific technology types and project conditions are required to provide accurate estimates of the usage of the water purifiers. If all six topics outlined are successfully completed the survey respondent can be classed as a user. If failure occurs in one or more of the topics, then the respondent is classed as a non-user. The project proponent needs to clearly define what is deemed a success or a failure for each topic area as part of its usage plan.

It is required that sufficient precautions are taken by enumerators to ensure that there is no bias while conducting the usage surveys. For example:

- The survey should not be conducted immediately after capacity building/awareness programs in the target households.
- Surveyors should not wear clothes with logos of their employers, etc.

Water quality testing - completed annually

The quality of the treated water will be assessed to ensure that it is fit for human consumption. Water quality test will be conducted based on section E of Annex 3 of *TPDDTEC* (Version 3.1). Accredited laboratory will be used to conduct the water quality test based on WHO standard of less than 1 Colony Forming Unit (CFU) of E.Coli/100 ml⁵⁰ and the Quality Standards for Sources of Domestic Water of The Environmental Management And Co-Ordination (Water Quality) Regulations, 2006⁵¹.

The laboratory used must demonstrate that it has an adequate quality management plan in place which addresses both quality assurance and quality control test procedures. Field testing kits also are eligible, e.g., based on Colony Forming Unit method or Most Probable Number method. To use the field testing kits the project shall meet the following requirements: Testing kits must be approved by national agency or meet standards set by relevant international organization e.g., US-EPA, or testing kits shall be tested for its accuracy and robustness prior to application for project level monitoring. Local or accredited laboratory shall conduct water quality tests using testing kits and a relevant ISO standard or an equivalent standard, in parallel with field testing kits. The water test results will be compared against values provided for different parameters under the manufacturer's specifications of the project technologies.

The water quality must be tested annually, with the first test shall be conducted at least after 6 months from the start date. The monitoring report may include:

- A full description of the water quality process which may be conducted either in the field or by transportation to laboratories.
- Endorsement by third parties of the testing.
- Justification of the appropriateness of the testing approach.

Leakage Assessment – completed every other year

As the leakage risks related to the safe water supply and treatment technologies can be deemed very low, the leakage can be ignored, and leakage is expected to be 0tCO₂e. Leakage investigation will be conducted every two years in line with section 3.8 of the applied GS methodology.

Hygiene campaigns and hygiene practice assessment - completed annually

⁵⁰ http://whqlibdoc.who.int/publications/2011/9789241548151_eng.pdf?ua=1

⁵¹ <http://extwprlegs1.fao.org/docs/pdf/ken84962.pdf>

The drinking water and hygiene practices will be assessed using the WHO/UNICEF Joint Monitoring Programme guidelines⁵² (JMP 2018). In-person or telephone or by messaging based survey including all the JMP core questions for drinking water and core questions for hygiene will be conducted.

B.7.3.5 Administering Data Collection and Quality Assurance and Quality Control

Administering Data Collection

The monitoring activities will be conducted by GS monitoring group. Then the survey result should be summarized by monitoring team member. The data will be reviewed by the project developer and VVB.

If the recorded raw data on the summary table of each parameter are reasonable and basically consistent with the actual state, the raw data will be archived. In addition, the raw questionnaires also should be preservation.

If the data record is missing or damaged during the monitoring periods, the following makeup process will be conducted:

- 1) The general principle is that conservative values will be used for the missing or damaged data. This is most conservative approach. The monitoring personnel will be trained before the starting of the project operation to ensure that each team member is fully aware of and able to strictly follow this conservative principle. During the monitoring process, the monitoring personnel will be required to strictly abide by the above conservative principle in data recording, i.e., use conservative values for all the missing or damaged data.
- 2) If this is due to the working error of the monitoring personnel, further train the person until he or she can perform the job properly. And in the meantime, use conservative value for the missing or damaged data.
- 3) If this is due to the inability or attitude of a particular worker in monitoring team, dismiss such worker and re-hire those with proper ability and attitude. And in the meantime, use conservative value for the missing or damaged data.
- 4) If some data recorded are significantly higher or lower than the normal range and inconsistent with actual situation, the monitoring personnel should ask for the reason. If the measurement is high or low due to the damage of measurement equipment, conservative value will be used. And need to calibrate or replace the measuring equipment immediately and avoid this situation in the future.

⁵² <https://washdata.org/report/jmp-2018-core-questions-household-surveys>

The PP shall perform statistical analysis on the data collated using Microsoft excel in an automated tool that will use standard deviation and mean values to determine outliers. If the monitoring results are satisfactory in terms of correct reporting, data completeness and correct analysis, the data will be accepted for the monitoring report.

QA/QC Procedures

Before implementing the project, the project owner will train the personnel of monitoring teams on how to properly conduct the monitoring process. The project proponent commits itself to accurate and transparent record keeping, monitoring and evaluation. All supporting documentation and records for the project must be easily accessible for spot checking and cross referencing by a third party; these records will be kept by Climate Neutral Kenya Limited. Contact information in the total sales database must allow a project auditor to easily contact and visit end users. An auditor must also be able to cross reference pertinent project documentation, which must include archives such as internal logs, financial accounts and sales records, as well as observations of retailer activities and sales performance.

SECTION C. DURATION AND CREDITING PERIOD

C.1. Duration of project

C.1.1. Start date of project

>>

09/10/2020, which is defined as the first purchase receipt for the water purifier sold to an end-user⁵³.

C.1.2. Expected operational lifetime of project

>>

15 years.

C.2. Crediting period of project

C.2.1. Start date of crediting period

>>

10/10/2020⁵⁴ or two years prior to the date of Project Design Certification, whichever is later.

C.2.2. Total length of crediting period

>>

15 years. As per section 4.1.5 of GS4GG Community Services Activity Requirements (Version 1.2), the crediting period is 5 years for each Design Certification Renewal Cycle and in total two Design Certification Renewal Cycles.

⁵³ During the usage rate survey, the purchase date and the start date to use of the water purifier were investigated. According to "summary table of Usage rate survey", among the users surveyed, there are users who used it on the day of purchase, so the start date of the project is the date of the water purifier sold, i.e., 09/10/2020.

⁵⁴ Although the project start date is 09/10/2020, which is the date of the first purchase receipt for the water purifier sold to an end-user, While, based on conservative considerations, in the calculation of emission reductions, it is considered that the water purifiers are used on the second day of purchase, so the start date of credit period is 10/10/2020.

SECTION D. SUMMARY OF SAFEGUARDING PRINCIPLES AND GENDER SENSITIVE ASSESSMENT

D.1. Safeguarding Principles that will be monitored

A completed Safeguarding Principles Assessment is in [Appendix 1](#), ongoing monitoring is summarized below.

Principles	Mitigation Measures added to the Monitoring Plan
Principle 1 Human Rights	Not required
Principle 2 Gender Equality and Women's Rights	Not required
Principle 3 Community Health, Safety and Working Conditions	The project is expected to have a positive effect on community health and safety. As a mitigation measure, the parameter of <i>Water Hygiene education campaigns</i> , Ongoing water quality indicated as the fraction of the samples that pass the test of bacterial quality to a standard of <1cfu/100ml ($M_{q,y}$) will be monitored during the crediting period.
Principle 4 Cultural Heritage, Indigenous Peoples, Displacement and Resettlement	Not required
Principle 5 Corruption	Not required
Principle 6 Economic Impacts	Not required
Principle 7.1 Emissions	Leakage emissions in project scenario were assessed and will be re-assessed every two years.
Principle 7.2 Energy supply	Not required
Principle 8 Water	Not required
Principle 9 Environment, Ecology and Land Use	Not required

D.2. Assessment that project complies with GS4GG Gender Sensitive requirements

Question 1 - Explain how the project reflects the key issues and requirements of Gender Sensitive design and implementation as outlined in the Gender Policy?	The project reflects the key gender issues and requirements of Gender Sensitive design and implementation. On the design side, the project is aimed to provide safe water for the local residents. The project will result in reduction of woody fuels that used to boil water in baseline scenario, which would generate harmful smoke and cause air pollution when burning in low efficiency and traditional stoves. In the overwhelming majority of the households in
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	Kenya, water fetching, fuel collection and purification activities are handled by women who are more exposed to the air pollution and the associated hazard. The availability of clean water in a reasonable distance is foreseen to reduce women's work load related to water purification, collection of fuel needed for boiling water and caring activities as the risk for water borne diseases will be reduced. Hence, largely women will benefit from the project activity. In order to ensure gender sensitive design and implementation, the project shall address any gender inequalities and gender-related risks that may be identified in project gender analyses. Gender-sensitive approaches in stakeholder consultation will also be ensured by promoting information sharing equitably with women and men stakeholders, which will be made available and presented in an accessible format across all stakeholder groups including those more marginalized. The approach will also include opportunities for stakeholders to share information in a two-way exchange, give regular feedback during implementation and ensure their views and priorities are incorporated in project design and implementation. On the implementation level, the project needs to hire local residents to carry out the whole project. When employing local residents, the project owner will not set up any barriers to the employment of women. The water purifier products will be used mainly by women, the project implementer opines those promotional activities are better addressed with women in the forefront. Women are able to be trained to master a new technology and establish awareness of health security. This woman prioritized mode of project development and implementation helps address gender equality issues; in the meantime, addressing issues related to environmental sustainability and natural resource management.
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Question 2 - Explain how the project aligns with existing country policies, strategies and best practices	The project does not involve and is not complicit in any form of discrimination based on gender, race, religion, sexual orientation or any other basis. The Kenya National Policy on Gender and Development (NPGD) spells out a policy approach of Gender mainstreaming and empowerment of women and clearly states that it is the right of women, men, girls and boys to participate in and benefit equally from the development process. The NPGD provides a framework for mainstreaming gender in all policies, planning and programming in Kenya and puts in place institutional mechanisms to ensure effective implementation ⁵⁵ . Vision 2030 Kenya's development blue print, aims to transform Kenya into a newly industrialized, middle-income country providing a high quality of life and secure environment for all its citizens by the year 2030. In the blueprint, gender imbalance is viewed as one of the key developmental challenges facing the country, and gender is prioritized in the social pillar. Vision 2030 aims to mainstream gender by making fundamental changes in four key areas, namely: opportunity; empowerment; capabilities; and vulnerabilities. It aims at increasing all round opportunities among women and increase the participation of women in all economic, social and political decision-making processes ⁵⁶ . The Third Medium Term Plan (MTP III) will implement the national policy on gender and development with an objective of increasing gender equality, empowerment of women, youth, and persons living with disability and other vulnerable groups⁵⁷. The project shall be implemented in the community where women are the primary beneficiary. Job opportunities will be provided fairly for women, which can help fostering their entrepreneurial skills and empower them be part in social dialogue.
Question 3 - Is an Expert required for the Gender Safeguarding Principles & Requirements?	No gender experts will be required for the Gender Safeguarding Principles & Requirements. The project shall neither

⁵⁵ Gender Policy, Ministry of Gender, Children and Social Development, Republic of Kenya. <https://land.igad.int/index.php/documents-1/countries/kenya/legislation-and-policies-3/policies-3/774-gender-policy/file>

⁵⁶ <https://vision2030.go.ke/>

⁵⁷ <http://www.fao.org/faolex/results/details/en/c/LEX-FAOC189643/>

	contributes to discrimination against women nor shall it reinforce gender-based discrimination and/or inequalities. Gender equality will be attained by ensuring that both men and women will have equal opportunities to jobs created and trainings. The project will further ensure equal pay for equal value of work.
Question 4 - Is an Expert required to assist with Gender issues at the Stakeholder Consultation?	No gender experts will be required to engage in the Stakeholder Consultation. The physical meeting held on 03/09/2021 was followed the <i>GS4GG Stakeholder Consultation and Engagement Requirements</i> (Version 1.2), i.e., the invitees of this meeting meet the gender guidelines, and minimum group of stakeholders to be consulted. The participants are (AB)End-user, (C) local authorities and (E) Local non-governmental organisations. All assessment questions related to safeguarding principles, including principle 2 "Gender Equality and Women's Rights", have been discussed during the stakeholder consultation meeting. Refer to Appendix 1 of the PDD for more details of the results for Safeguarding Principal Assessment.

SECTION E. SUMMARY OF LOCAL STAKEHOLDER CONSULTATION

The below is a summary of the 2 step GS4GG Consultation for monitoring purposes. Please refer to the separate Stakeholder Consultation Report for a complete report on the initial consultation and stakeholder feedback round.

E.1. Summary of stakeholder mitigation measures

>>

Local stakeholder consultation of the project was organized in line with the GS4GG stakeholder Procedure Requirements and Guidelines.

The project owner invited the stakeholders to participate in the stakeholder consultation meeting on the project through email and public notice. In order to facilitate the invitees to understand the purpose of the meeting and relevant information of the project, the project owner briefly introduced the key information of the project during the invitation. The project owner invited 13 people to participate the physical meeting through email and invited more end-user to participate

the physical meeting through the public notice. Moreover, the Project owner tried to invite the same number of males and females.

21 people, including local residents and experts from local government attended in the 1st round of Local stakeholder consultation physical meeting held on 03/09/2021 in green lawn⁵⁸ nearby the Wote town office, along the wote-makindu road, Makueni County, Kenya. In the meeting, the project owner introduced the purpose and significance of this meeting project activity and Explanation of the project in non-technical terms. After the description of the project, stakeholders conducted the question-and-answer session about the project. Moreover, in the meeting, the any potential risks and positive impacts of the project, the monitoring plan for sustainable development impacts and the continuous input & Grievance Mechanism were also be discussed.

The comments received during the stakeholder consultation meeting can summarized below:

Comment: How does the water filter work?

Simply pour untreated water into the water filter. Water seeps through the ceramic filter, removing bacteria and other suspended solids, and treated water is stored in a large plastic receptacle, which is easily accessed by a spigot.

Comment: Is the water treated by the filter safe enough for drinking?

Ceramic filters will filter out bacteria and other suspended solids to make the water safe. Filtered water is safe drinking water that has been tested. We will regularly test whether the water quality meets the drinking requirements.

Comment: How does the project produce carbon emissions?

By using ceramic filters, the users do not need any fuel to boil water for drinking. The carbon emissions produced by firewood and charcoal firing is reduced in the project scenario.

Comment: How do you monitor the emission reductions since it is invisible?

We do not monitor the emission directly. By monitoring the reduced quantity of firewood or charcoal, we can calculate the emission reductions. The calculation sheet is provided if you are interested.

Comment: How can we benefit from the project implementation?

Apart from the cheap and clean drinking water, the stakeholders can also benefit from the project by ensuring the maintenance and monitoring of the project operation. The project owner needs local employees to sell water filters and monitor the parameters related to the emission calculation.

⁵⁸ In the Email of invitation to the physical meeting, the meeting place is Wote town office, along the wote-makindu road, Makueni County, Kenya. While, affected by the epidemic, in order to comply with local epidemic prevention requirements, the meeting must be held outdoors, so the final meeting place is green lawn nearby the Wote town office, along the wote-makindu road, Makueni County, Kenya.

The 2nd round of Stakeholder Consultation, the Stakeholder feedback round is held for two months between 03/08/2021 and 02/10/2021. All stakeholders that participated in the Stakeholder Consultation physical meeting were invited to the Stakeholder Feedback Round, including those who were invited but unable to attend. The electronic version and paper copy Key Project Information and PDD are sent to the stakeholders for review and feedback. People can give feedback by replying the invitation email, telephone etc., and all the feedback will be recorded. No stakeholder feedback was received during the Stakeholder Feedback Round.

In August 2021, the project owner invited and consulted relevant experts, including the staff of relevant institutions and organizations with professional knowledge related to project information and safeguarding requirements, to ensure that the implementation of the project meets the requirements of GS4GG Requirements, and will not affect the rights of local residents or local ecology and hydrology while bringing benefits to local residents. On the aspects of human rights, gender equality and women's rights, community health, safety and working conditions, cultural heritage, indigenous people, displacement and resettlement, corruption, economic impacts, climate and energy, water, and environment, ecology and land use, the project owner consulted the experts of the ministry of environment, ministry of gender and youth, Federation of Women Law and Women's Federation of Nanyang City mainly through email. The project information and Safeguarding principles are provided to the experts. These experts affirmed the practice of creating jobs for the local area and encouraging the recruitment of women in the project, and also affirmed work in the project will be no negative impact in these aspects during the implementation of the project, and there is no improvement suggestion on the design of the project at the current stage.

E.2. Final continuous input / grievance mechanism

Method	Include all details of Chosen Method (s) so that they may be understood and, where relevant, used by readers.
Continuous Input / Grievance Expression Process Book (mandatory)	The grievance expression book is kept by the chief officer Environment, Energy & Natural Resources of Makueni County, Trans Nzoia County, Kirinyaga County, Bungoma County and Nakuru County. All stakeholders are allowed to record their grievances or comments in the book. The project owner will assign specific person in each sub county to deal with related issues. When assessing the relevance and appropriateness of the stakeholders' comments, the project owner will apply a gender lens.
GS Contact (mandatory)	help@goldstandard.org

Other	Feedback can be made directly through visiting Climate Neutral Kenya Limited or through e-mail mosesatemba8810@gmail.com
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APPENDIX 1 - SAFEGUARDING PRINCIPLES ASSESSMENT

Complete the Assessment below and copy all Mitigation Measures for each Principle into SECTION D above. Please refer to the instructions in the [Guide to Completing](#) this Form below.

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
Principle 1. Human Rights			
1. The Project Developer and the Project shall respect internationally proclaimed human rights and shall not be complicit in violence or human rights abuses of any kind as defined in the Universal Declaration of Human Rights 2. The Project shall not discriminate with regards to participation and inclusion	No	To live in a clean and healthy environment is the fundamental right to all citizens in Kenya. The project pursues to safeguard this fundamental right by availing the clean and safe water solutions to the households. The fundamental right is preserved. All households in the project region that respect the principles and values of sustainable development can equally participate and benefit from the project. The project is implemented under the laws of Kenya. There is no risk of violence or abuse of human rights in this project. The project	Not required

		does not discriminate on gender, race, religion, sexual orientation or any other aspect.	
Principle 2. Gender Equality and Women's Rights			
1. The Project shall not directly or indirectly lead to/contribute to adverse impacts on gender equality and/or the situation of women 2. Projects shall apply the principles of nondiscrimination, equal treatment, and equal pay for equal work 3. The Project shall refer to the country's national gender strategy or equivalent national commitment to aid in assessing gender risks 4. (where required) Summary of opinions and recommendations of an Expert Stakeholder(s)	No	The project targets the most vulnerable people in the country to ease access to clean and safe water. The project will not directly or indirectly reinforce gender-based discrimination and shall not lead to/contribute to adverse impacts on gender equality and/or the situation of women. In fact, the project increases women's access to or control of resources, entitlements and benefits by providing easy access to clean water and equal job opportunities. The project will not set up any barriers to the employment of women. All employees have benefits based on pregnancy, maternity/paternity leave, or marital status according to the <i>Employment Act, the Labour Relations Act, the Labour Institutions Act, the Work Injury</i>	Not required.

		<i>Benefits Act, and the Occupational Health and Safety Act.</i> Therefore, the safeguarding principle related to Gender Equality and Women's Rights is not triggered during the project design and implementation.	
Principle 3. Community Health, Safety and Working Conditions			
1. The Project shall avoid community exposure to increased health risks and shall not adversely affect the health of the workers and the community	No	The project reduces the health risks caused by traditional water treatment method or direct use of untreated water. The water purifiers used in the project activity will not cause community exposure to increased health risks and shall not adversely affect the health of the workers and the community.	The project is expected to have a positive effect on community health and safety. As a mitigation measure, the parameter of <i>Water Hygiene education campaigns</i> , Ongoing water quality indicated as the fraction of the samples that pass the test of bacterial quality to a standard of <1cfu/100ml ($M_{q,y}$) will be monitored during the crediting period.
Principle 4.1 Sites of Cultural and Historical Heritage			
Does the Project Area include sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture?	No	The water purifier is a small household device. It is not used in sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture.	Not required
>>		The project does not utilise Cultural Heritage, including the knowledge, innovations, or	

		practices of local communities, affected communities.	
Principle 4.2 Forced Eviction and Displacement			
Does the Project require or cause the physical or economic relocation of peoples (temporary or permanent, full or partial)?	No	The water purifier used in the project activity is a small household equipment. It obviously does not cause physical or economic relocation of peoples.	Not required
>>			
Principle 4.3 Land Tenure and Other Rights			
Does the Project require any change, or have any uncertainties related to land tenure arrangements and/or access rights, usage rights or land ownership?	No	The water purifier used in the project activity is a small household equipment. It can be used at the users’ home. The project does not require any change to land tenure arrangements and/or other rights such as resource access rights, community-based property rights and customary rights.	Not required
>>			
Principle 4.4 Indigenous Peoples			
1. The Project Developer shall identify all communities of Indigenous Peoples within the Project area of influence who may be affected directly or indirectly by the Project.	No	People have the same and equal access to the clean and safe water, and no one will be affected directly or indirectly in a negative way by the project. In addition, there are no indigenous people present within	Not required

2. The Project Developer shall recognise and respect the indigenous people's collective rights to own, use, and develop and control the lands, resources and territories that they have traditionally owned, occupied or otherwise used or acquired, including lands and territories for which they do not yet possess title. 3. The Project Developer shall respect, protect, conserve and shall not take the cultural, intellectual, religious and spiritual property of indigenous peoples without their free, prior and informed consent (FPIC). 4. The Project Developer shall ensure that the indigenous people are provided with the equitable sharing of benefits to be derived from utilisation and/or		the area of influence nor the project is located on territory claimed by indigenous people.	
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commercial development of natural resources on lands and territories or use of their traditional knowledge and practices by the Project. This shall be done in a manner that is culturally appropriate and inclusive and that does not impede land rights or equal access to basic services including health services, clean water, energy, education, safe and decent working conditions and housing. 5. The opinions and recommendations of an Expert Stakeholder(s) shall be sought and demonstrated as being included in the project design.			
>>			
Principle 5. Corruption			
6. The Project shall not involve, be complicit in or inadvertently contribute to or reinforce corruption or corrupt Projects	No	The project is implemented on the ground by the social enterprise. The ethical codes of the project partners are against corruption. In addition, the companies	Not required

		comply with the <i>Anti-Corruption and Economic Crimes Act (ACECA) and the Public Officer Ethics Act.</i> Hence, the Project does not involve, be complicit in or inadvertently contribute to or reinforce corruption or corrupt Projects.	
Principle 6.1 Labour Rights			
1. The Project Developer shall ensure that all employment is in compliance with national labour occupational health and safety laws and with the principles and standards embodied in the ILO fundamental conventions 2. Workers shall be able to establish and join labour organisations 3. Working agreements with all individual workers shall be documented and implemented and include: a) Working hours (must not exceed 48 hours per week on a regular basis), AND	No	The project complies with <i>the Employment Act, the Labour Relations Act, the Labour Institutions Act, the Work Injury Benefits Act, and the Occupational Health and Safety Act.</i> The project owner will sign contracts with employees. The labor contracts specify working hours, tasks and payments. All employees have benefits based on social security, pregnancy, maternity/paternity leave, or marital status according to the labor laws of Kenya. Besides, the employees also have the right to establish labour unions and to carry on labour union activities.	Not required

b) Duties and tasks, AND c) Remuneration (must include provision for payment of overtime), AND d) Modalities on health insurance, AND e) Modalities on termination of the contract with provision for voluntary resignation by employee, AND f) Provision for annual leave of not less than 10 days per year, not including sick and casual leave. 4. No child labour is allowed (Exceptions for children working on their families' property requires an <u>Expert Stakeholder</u> opinion) 5. The Project Developer shall ensure the use of appropriate equipment, training of workers, documentation and reporting of accidents and incidents, and emergency preparedness and response measures		All employees would provide their age information document, e.g. ID, when signing the labour contract, and the project owner did not and will not employ any child labour.	
Principle 6.2 Negative Economic Consequences			

1. Does the project cause negative economic consequences during and after project implementation?	No	The equipment procurement cost of the project will be borne by the project owner, and part or all of the cost will be recovered by selling equipment to residents. In addition, the project owner will bear the cost of equipment maintenance, project certification fee and the cost of the project monitoring personnel. The project will be economically feasible through the sale of emission reduction credits.	Not required
>>			
Principle 7.1 Emissions			
Will the Project increase greenhouse gas emissions over the Baseline Scenario?	No	The project uses zero energy water purifier for water treatment, which does not generate any emissions. Therefore, the does not lead to an increase in greenhouse gas emissions over the baseline scenario.	Leakage emissions in project scenario will be monitored and reported before first verification and every two years.
>>			
Principle 7.2 Energy Supply			
Will the Project use energy from a local grid or power supply (i.e., not connected to a national or regional grid) or fuel resource (such as wood, biomass) that provides for other local users?	No	The project uses zero energy water purifier for water treatment, and will not use energy from the local grid or power supply or fuel resource	Not required

>>		supply that provides for other local users.	
Principle 8.1 Impact on Natural Water Patterns/Flows			
Will the Project affect the natural or pre-existing pattern of watercourses, ground-water and/or the watershed(s) such as high seasonal flow variability, flooding potential, lack of aquatic connectivity or water scarcity?	No	The project is not involved in abstraction from water resources required to support biodiversity and other ecosystem services. It only changes the water treatment method instead of water resources, and will not negatively affect the natural or pre-existing pattern of watercourses, groundwater and/or the watershed(s).	Not required
>>			
Principle 8.2 Erosion and/or Water Body Instability			
Could the Project directly or indirectly cause additional erosion and/or water body instability or disrupt the natural pattern of erosion?	No	The water purifiers used at the residents' homes in the project could not directly or indirectly cause additional erosion and/or water body instability or disrupt the natural pattern of erosion, and could not directly or indirectly impact on surface and ground waters or soil erosion on slopes.	Not required
>>			
Principle 9.1 Landscape Modification and Soil			
Does the Project involve the use of land and soil for production of crops or other products?	No	The project does not involve the production, harvesting, and/or management of living natural	Not required

>>		resources by small-scale landholders and/or local communities. The water purifiers used at the residents' homes in the project does not involve any use of land or soil and will not cause degradation in existing landscape function and services. It will not affect the health condition of any soils.	
Principle 9.2 Vulnerability to Natural Disaster			
Will the Project be susceptible to or lead to increased vulnerability to wind, earthquakes, subsidence, landslides, erosion, flooding, drought or other extreme climatic conditions?	No	The water purifiers of the project are used at the residents' homes and does not involve any land use changes. It would not lead to the exacerbation of impacts caused by natural or man-made hazards, such as landslides or floods.	Nor required
>>			
Principle 9.3 Genetic Resources			
Could the Project be negatively impacted by or involve genetically modified organisms or GMOs (e.g., contamination, collection and/or harvesting, commercial development, or take place in facilities or farms that include	No	The project does not involve the use of GMOs, so it will not have negatively impacted by the use of genetically modified organisms or GMOs.	Nor required

GMOs in their processes and production)?			
>>			
Principle 9.4 Release of pollutants			
Could the Project potentially result in the release of pollutants to the environment?	No	The water purifiers used in this project will not produce any emissions and releasing no pollutants to the environment.	Not required
>>			
Principle 9.5 Hazardous and Non-hazardous Waste			
Will the Project involve the manufacture, trade, release, and/or use of hazardous and non-hazardous chemicals and/or materials?	No	The project does not involve chemicals or materials subject to international bans or phase-outs. It does not involve the manufacture, trade, release, and/or use of hazardous and non-hazardous chemicals and/or materials.	Not required
>>			
Principle 9.6 Pesticides & Fertilisers			
Will the Project involve the application of pesticides and/or fertilisers?	No	The project does not involve any application of pesticides and/or fertilizers.	Not required
>>			
Principle 9.7 Harvesting of Forests			
Will the Project involve the harvesting of forests?	No	The water purifiers of the project are used at the residents' homes	Not required
>>			

		project, and the project does not involve the harvesting of forests.	
Principle 9.8 Food			
Does the Project modify the quantity or nutritional quality of food available such as through crop regime alteration or export or economic incentives?	No	The water purifiers of the project are used at the residents' homes, and the project does not involve the modification of the quantity or nutritional quality of food available.	Not required
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Principle 9.9 Animal husbandry			
Will the Project involve animal husbandry?	No	The water purifiers of the project are used at the residents' homes, and the project does not involve animal husbandry.	Not required
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Principle 9.10 High Conservation Value Areas and Critical Habitats			
Does the Project physically affect or alter largely intact or High Conservation Value (HCV) ecosystems, critical habitats, landscapes, key biodiversity areas or sites identified?	No	The project is not located in an area within a high conservation value area or within critical natural habitats. The water purifiers of the project are used at the residents' homes, and the project does not physically affect or alter largely intact or High Conservation Value (HCV) ecosystems, critical habitats, landscapes, key biodiversity areas or sites identified.	Not required
>>			
Principle 9.11 Endangered Species			

Are there any endangered species identified as potentially being present within the Project boundary (including those that may route through the area)? AND/OR Does the Project potentially impact other areas where endangered species may be present through transboundary affects?	No	There are no endangered species identified as potentially being present within the Project boundary. In addition, the water purifiers used in the project will not pose a threat to any species.	Not required
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APPENDIX 2- CONTACT INFORMATION OF PROJECT PARTICIPANTS

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APPENDIX 3- LUF ADDITIONAL INFORMATION

Not applicable.

APPENDIX 4-SUMMARY OF APPROVED DESIGN CHANGES

Not applicable.

Revision History

Version	Date	Remarks
1.2	14 October 2020	Hyperlinked section summary to enable quick access to key sections Improved clarity on Key Project Information Inclusion criteria table added Gender sensitive requirements added Prior consideration (1 yr rule) and Ongoing Financial Need added Safeguard Principles Assessment as annex and a new section to include applicable safeguards for clarity Improved Clarity on SDG contribution/SDG Impact term used throughout Clarity on Stakeholder Consultation information required Provision of an accompanying Guide to help the user understand detailed rules and requirements
1.1	24 August 2017	Updated to include section A.8 on 'gender sensitive' requirements
1.0	10 July 2017	Initial adoption